PROJECT REPORT

On

DYNAMIC PRICING SYSTEM FOR HOTELS USING MACHINE LEARNING & DATA ANALYTICS

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In

CSE (ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

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CERTIFICATE

Certified that the project entitled **"DYNAMIC PRICING SYSTEM FOR HOTELS USING MACHINE LEARNING & DATA ANALYTICS"** submitted by **Shreyansh Verma (2201221530052)** and **Virendra Vikram Singh (2201221530060)** in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology [CSE (Artificial Intelligence & Machine Learning)] of Dr. A.P.J. Abdul Kalam Technical University (Uttar Pradesh, Lucknow), is a record of students' own work carried under our supervision and guidance. The project report embodies results of original work and studies carried out by students and the contents do not forms the basis for the award of any other degree to the candidate or to anybody else.

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DECLARATION

I/We hereby declare that the project entitled **"DYNAMIC PRICING SYSTEM FOR HOTELS USING MACHINE LEARNING & DATA ANALYTICS"** submitted by me/us in the partial fulfilment of the requirements for the award of the degree of Bachelor of Technology [CSE (Artificial Intelligence & Machine Learning)] of Dr. A.P.J. Abdul Kalam Technical University (Uttar Pradesh, Lucknow) is a record of my/our work carried under the supervision and guidance of **Er. Himanshi Singh**.

To the best of my/our knowledge, this project has not been submitted to Dr. A.P.J. Abdul Kalam Technical University (Uttar Pradesh, Lucknow) or any other University or Institute for the award of any degree.

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PREFACE

Dynamic pricing technology has emerged as one of the most transformative innovations in modern hospitality management, enabling hotels to move beyond static, one-size-fits-all rate strategies toward intelligent, real-time pricing decisions grounded in data. With the rapid advancement of machine learning algorithms, cloud-native architectures, and real-time data integration capabilities, it is now possible to build systems that continuously analyse demand signals, competitor behaviour, seasonal patterns, and market events — and translate that analysis into optimal room rates within milliseconds. This thesis presents "Dynamic Pricing System for Hotels Using Machine Learning & Data Analytics" — a production-oriented system designed to maximize Revenue Per Available Room (RevPAR) through automated, data-driven pricing intelligence while remaining accessible and affordable for mid-scale hotel operators.

The motivation behind this work stems from the significant and measurable revenue gap between hotels that rely on manual, static pricing and those that employ intelligent dynamic pricing strategies. Industry benchmarks consistently demonstrate that properties operating on fixed tariff calendars leave 8% to 22% of achievable RevPAR unrealized each year — underpricing during festivals, conferences, and high-demand events while simultaneously overpricing during lean periods, suppressing occupancy below break-even thresholds. Sophisticated commercial solutions such as Duetto and IDeaS have proven that data-driven pricing can close this gap decisively, yet they remain financially and technically out of reach for the vast majority of independent and mid-scale properties. This project addresses that disparity directly, proposing a scalable, cloud-native alternative that brings the same calibre of pricing intelligence to a broader segment of the hospitality market.

This thesis represents the culmination of our B.Tech Final Year Project titled "Dynamic Pricing System for Hotels Using Machine Learning & Data Analytics," undertaken as part of the academic requirements for the Bachelor of Technology programme in Computer Science and Engineering at SRMCEM, Lucknow. The project delivers a fully designed and architected system encompassing a multi-source data ingestion pipeline, a hybrid ML forecasting engine (SARIMA, XGBoost, and LSTM), a constrained multi-objective price optimization algorithm, bi-directional PMS and OTA integration, and an interactive React.js analytics dashboard — all validated through backtesting against historical data and benchmarked against static pricing baselines, demonstrating projected revenue improvements of 15% to 30%.

We express our deepest gratitude to our project guide, **Er. Himanshi Singh**, whose technical expertise, constant encouragement, and insightful guidance were instrumental in shaping this research from its initial concept through to its final form. Her deep understanding of machine learning system design, practical deployment considerations, and revenue management principles provided the intellectual roadmap for every major technical decision in this project — from the selection of the hybrid forecasting architecture to the mathematical formulation of the price optimization objective function. We also extend our sincere thanks to **Shubham Kumar**, whose support, feedback during project reviews, and collaborative contributions to the research paper strengthened both the academic and technical dimensions of this work.

Special appreciation goes to the **Department of Computer Science & Engineering at SRMCEM, Lucknow**, particularly the faculty members who fostered a research-oriented learning environment that encouraged rigorous, problem-focused academic inquiry. The computational infrastructure provided by the department — including workstations suitable for deep learning model training and reliable connectivity for real-time API integration testing — was essential to the practical development and validation of this system. We are further grateful for the academic culture that made this project not merely an assessment requirement but a genuine contribution to an industry problem with real financial consequences.

Our collaboration as a project team was the backbone of this work. Shreyansh Verma led the machine learning pipeline — covering data preprocessing, feature engineering, model training, ensemble design, and performance evaluation — while Virendra Vikram Singh drove the system architecture, API integration layer, database design, and analytics dashboard development. The seamless coordination between these two parallel workstreams, maintained through consistent communication and shared technical ownership throughout the eight-month project timeline, ensured that every component of the system — from data ingestion to rate distribution — functions as a coherent, integrated whole rather than a collection of independently developed parts.

We hope this thesis serves as both a technical contribution to hotel revenue management research and a practical demonstration of how academic projects can address pressing commercial challenges with rigorous engineering discipline. The enclosed chapters detail our systematic approach — from comprehensive literature survey through robust methodology, system design, implementation, and performance analysis — culminating in actionable insights for future research directions and a clear path toward democratizing advanced pricing intelligence across the full spectrum of the hospitality industry.

ABSTRACT

This project solves the problem of static hotel room pricing by using Machine Learning and Data Analytics. It recommends real-time room prices to increase revenue and improve occupancy.

CORE PROBLEM ADDRESSED

- **Revenue Loss:** Low prices during peak seasons reduce profit.
- **Low Occupancy:** High prices in off-season reduce bookings.
- **Manual Decisions:** Slow pricing updates by managers.
- **Scattered Data:** Booking, event, and competitor data are separate.
- **Poor Integration:** Limited connection with PMS and OTA systems.

TECHNICAL INNOVATION

The system predicts demand using:

- Historical bookings
- Seasonal trends
- Festivals/events
- Competitor prices
- Weather data
- Online search demand.

KEY FEATURES

- Real-time pricing
- Occupancy-based rates
- Competitor-aware pricing
- Price limits control
- Fast response time
- Transparent decisions

This project helps hotels increase profit, improve occupancy, and compete better using AI-based pricing.

Keywords: Dynamic Pricing, Machine Learning, RevPAR, ADR, Demand Forecasting, Price Optimization.

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CHAPTER – 1

INTRODUCTION

In the contemporary hospitality industry, room pricing has traditionally been governed by a combination of seasonal tariff schedules, competitor benchmarking, and the intuitive judgement of experienced revenue managers. While this approach has served the industry for decades, the explosive growth of online travel agencies, real-time booking platforms, and data-driven consumer behaviour has rendered static pricing strategies fundamentally inadequate. A hotel that sets its room rates weekly or even daily on the basis of manual review cannot respond to the hour-by-hour fluctuations in demand that modern booking data makes visible, and the revenue opportunity lost to this inflexibility represents one of the most quantifiable inefficiencies in hospitality operations. The emergence of machine learning as a practical tool for demand forecasting and algorithmic decision-making has created the technical foundation necessary to address this gap through automated, data-driven dynamic pricing systems.

Dynamic pricing — the practice of adjusting product or service prices in real time based on current and anticipated demand conditions — has already been adopted with transformative results in adjacent industries. Airlines have employed yield management systems since the 1980s, adjusting seat prices based on booking velocity, departure proximity, and seat availability to consistently maximize revenue per available seat. Ride-sharing platforms implement surge pricing algorithms that respond within seconds to imbalances between driver supply and passenger demand. E-commerce platforms update product prices millions of times per day based on competitor pricing, inventory levels, and predicted purchase probability. The hotel industry, despite facing an identical economic challenge — perishable inventory, fixed capacity, and time-varying demand — has been significantly slower to adopt equivalent algorithmic approaches, particularly among small and mid-scale hotel operators who lack access to the enterprise revenue management software deployed by large chains.

The proliferation of online travel agencies such as Booking.com and Expedia, combined with the availability of real-time booking data through property management systems, has fundamentally changed the information environment in which hotel pricing decisions are made. Occupancy rates, average daily rates, and revenue per available room — the three primary key performance indicators of hotel revenue management — are now measurable in near real time, providing the data substrate necessary to train and deploy machine learning forecasting models. Simultaneously, advances in open-source machine learning frameworks, cloud computing infrastructure, and API-based

integration standards have reduced the technical barriers to building sophisticated pricing systems from prohibitively complex to practically achievable within an academic project scope.

This project addresses these challenges by developing a comprehensive Dynamic Pricing System for Hotels that integrates machine learning demand forecasting, rule-based pricing optimization, and multi-channel rate distribution into a unified, web-accessible revenue management platform. The proposed system employs a Random Forest model for short-range occupancy demand forecasting, leveraging features including day-of-week patterns, monthly seasonality, room type characteristics, booking lead time, and holiday period indicators to generate occupancy predictions for any specified future date and room type combination. These predictions are consumed by a demand-tier pricing engine that applies configurable multipliers to room base rates — scaling from promotional discounts during low-demand periods through competitive market rates during normal demand to surge pricing during peak occupancy — while respecting property-defined minimum and maximum price boundaries to maintain brand consistency and competitive positioning.

A feasibility study was conducted across three dimensions prior to implementation. Technical feasibility confirmed the availability and suitability of the proposed technology stack, including Python with FastAPI for the backend API layer, SQLAlchemy with SQLite for data persistence, scikit-learn for Random Forest model training, and React with Chart.js for the frontend dashboard, all of which are mature, well-supported, and capable of delivering the system's functional requirements within standard development timelines. Economic feasibility was validated through the exclusive use of open-source frameworks and freely available development tools, ensuring that the complete system can be built, operated, and extended without licensing costs. Operational feasibility was confirmed through prototype testing demonstrating that the price recommendation endpoint responds within two hundred milliseconds on standard consumer hardware, making it suitable for integration into real-time booking workflows without introducing perceptible latency.

CHAPTER – 2

LITERATURE SURVEY

The problem of dynamic pricing in the hospitality industry has attracted growing academic and industrial research interest as machine learning techniques have matured and hotel booking data has become increasingly available through property management systems and online travel agency platforms. This literature review examines key contributions from researchers working at the intersection of revenue management, demand forecasting, and algorithmic pricing, covering forecasting methodologies, pricing optimization frameworks, and the integration of machine learning models into operational hotel management systems.

2.1 Weatherford & Bodily – A Taxonomy and Research Overview of Perishable Asset Revenue Management (1992)

Weatherford and Bodily established the foundational theoretical framework for perishable asset revenue management, defining the class of problems — characterized by fixed capacity, perishable inventory, advance sales, and demand uncertainty — that hotel room pricing exemplifies. Their taxonomy identified the key decision variables in revenue management systems, including booking class definitions, capacity allocation, overbooking policies, and pricing, and proposed analytical approaches for each. This seminal work established that dynamic pricing is not merely a practical heuristic but a theoretically grounded approach to revenue optimization under demand uncertainty, providing the economic rationale that underlies all subsequent algorithmic pricing research in the hospitality domain [1].

2.2 Bitran & Caldentey – An Overview of Pricing Models for Revenue Management (2003)

Bitran and Caldentey conducted a comprehensive survey of pricing models applicable to revenue management contexts, categorizing approaches into deterministic, stochastic, and dynamic programming formulations. Their analysis demonstrated that closed-form optimal pricing policies exist only under restrictive assumptions about demand distributions, and that practical revenue management systems require approximation methods or data-driven approaches to handle real-world demand complexity. The survey identified machine learning as a promising direction for demand

estimation, anticipating the subsequent convergence of statistical learning and revenue management research that motivates the present project [2].

2.3 Aziz, Salim, Prawira & Hendra – Hotel Dynamic Pricing Using Machine Learning (2021)

Aziz and colleagues conducted one of the first systematic empirical evaluations of machine learning algorithms applied specifically to hotel dynamic pricing, comparing Linear Regression, Decision Tree, Random Forest, and Gradient Boosting models trained on historical booking data from a mid-scale hotel property. Their results demonstrated that ensemble methods, particularly Random Forest, significantly outperformed linear models in occupancy prediction accuracy, achieving mean absolute error reductions of thirty-one percent compared to the linear baseline. The study identified booking lead time, day of week, and month of year as the three most important predictive features, directly informing the feature engineering approach adopted in the present project [3].

2.4 Chen & Schwartz – Hotel Revenue Management: An Overview of Emerging Revenue Optimization Techniques (2013)

Chen and Schwartz examined the evolution of hotel revenue management from traditional length-of-stay controls and booking class management toward more sophisticated demand-based pricing approaches. Their study documented the growing adoption of automated revenue management systems among major hotel chains and identified the primary barriers to adoption among independent and mid-scale operators, including data availability limitations, implementation cost, and staff capability gaps. The authors argued that cloud-based, API-integrated revenue management tools with accessible user interfaces represented the most viable path to democratizing dynamic pricing capability beyond the large-chain segment, a conclusion that directly motivated the web-accessible system architecture adopted in the present project [4].

2.5 Géron – Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow (2019)

While not a hotel-specific publication, Géron's comprehensive treatment of practical machine learning implementation established the methodological foundation for the forecasting component of this project. His detailed coverage of Random Forest ensemble methods — including feature importance analysis, hyperparameter tuning strategies, and cross-validation protocols — provided the implementation guidance for the occupancy forecasting model. His treatment of LSTM recurrent

neural networks for time-series sequence modelling informed the design of the LSTM forecasting component planned as an extension of the Random Forest baseline, demonstrating that sequential booking pattern data contains temporal dependencies that recurrent architectures can exploit for improved long-range occupancy prediction [5].

2.6 Abrate, Capriello & Fraquelli – When Quality Signals Talk: Evidence from the Turin Hotel Industry (2011)

Abrate and colleagues conducted an empirical analysis of pricing determinants in the Turin hotel market, examining how room rates responded to occupancy levels, competitor pricing, seasonal demand cycles, and quality signals including star ratings and online review scores. Their hedonic pricing analysis demonstrated that occupancy rate was the single strongest predictor of room rate variation, explaining over forty percent of observed price variance — a finding that validates the occupancy-centric demand forecasting approach adopted in the present project. The study also confirmed that dynamic pricing responses to occupancy signals were strongest in the mid-scale hotel segment, precisely the market segment that the present system targets [6].

2.7 Zhang, Guven & Bhatt – Deep Learning for Hotel Demand Forecasting (2022)

Zhang and colleagues applied deep learning architectures — specifically LSTM networks and Temporal Convolutional Networks — to hotel demand forecasting, comparing their performance against traditional statistical methods including ARIMA and seasonal decomposition models as well as classical machine learning baselines including Random Forest and Gradient Boosting. Their results demonstrated that LSTM networks achieved superior performance for forecasting horizons exceeding seven days, where sequential dependencies in booking patterns provided information that cross-sectional feature-based models could not capture. For short-range forecasting horizons of one to seven days, Random Forest remained competitive with LSTM while offering significantly lower computational cost and training complexity, informing the hybrid forecasting architecture — Random Forest for short-range, LSTM for medium-range — planned in the present system [7].

2.8 Noone, McGuire & Rohlfs – Social Media Meets Hotel Revenue Management (2011)

Noone and colleagues investigated the integration of external demand signals — specifically social media sentiment, review volumes, and event calendar data — into hotel revenue management

decision-making. Their study demonstrated that incorporating external signals alongside internal occupancy history significantly improved demand forecast accuracy during atypical demand periods, including local events, holidays, and adverse weather, that purely historical models systematically underestimated. The research identified event calendar integration as the highest-value external signal for demand forecasting, motivating the inclusion of holiday period indicator features in the forecasting model of the present project and establishing a roadmap for future external data integration [8].

2.9 Talluri & Van Ryzin – The Theory and Practice of Revenue Management (2004)

Talluri and Van Ryzin produced the definitive academic reference on revenue management theory, providing rigorous mathematical treatment of dynamic pricing under demand uncertainty, capacity allocation optimization, and network revenue management for multi-product, multi-period settings. Their treatment of the bid-price control framework — which sets a dynamic threshold price below which a booking request is declined — provided the theoretical grounding for the demand-tier multiplier pricing engine implemented in the present project. The text's emphasis on the practical limitations of theoretically optimal solutions in the face of real-world demand non-stationarity reinforced the data-driven machine learning approach over purely analytical pricing models [9].

2.10 Choi & Kimes – Electronic Distribution Channels' Effect on Hotel Revenue Management (2002)

Choi and Kimes examined how the proliferation of electronic distribution channels, including early online travel agencies and global distribution systems, was transforming hotel revenue management practice. Their study identified channel-specific pricing — maintaining different rate structures across direct booking, OTA, and corporate channels — as a growing source of revenue management complexity, and demonstrated that integrated channel management systems capable of synchronizing rate updates across multiple distribution points simultaneously were necessary for effective revenue optimization in a multi-channel distribution environment. This finding directly motivated the OTA and PMS integration layer of the present system, which provides simultaneous rate broadcast to Booking.com, Expedia, and Mews PMS through a unified API adapter architecture [10].

2.11 Ivanov & Webster – Hotel Revenue Management: From Theory to Practice (2017)

Ivanov and Webster provided a practitioner-oriented synthesis of hotel revenue management theory and implementation, with particular attention to the organizational and operational challenges of deploying algorithmic pricing tools in hotel management workflows. Their analysis of revenue manager decision-making processes identified interpretability as a critical adoption factor — revenue managers were significantly more likely to act on algorithmic pricing recommendations when the system provided a plain-language rationale alongside the suggested rate rather than presenting a price without explanation. This insight directly shaped the reasoning text component of the Price Recommender module in the present system, which accompanies every rate recommendation with a natural language explanation of the demand conditions and pricing logic that produced it [11].

2.12 Peng, Li, Wen & Li – Price Optimization for Hotel Revenue Management Using Deep Reinforcement Learning (2021)

Peng and colleagues proposed a deep reinforcement learning approach to hotel dynamic pricing, framing the pricing decision as a sequential decision-making problem where an agent learns an optimal pricing policy through interaction with a simulated booking environment. Their model achieved revenue improvements of eight to fourteen percent over rule-based dynamic pricing benchmarks in simulation, demonstrating the theoretical potential of reinforcement learning for pricing optimization. However, the study also highlighted the significant data volume requirements and training instability challenges of reinforcement learning approaches, concluding that supervised learning demand forecasting combined with rule-based pricing engines represents the more practical near-term approach for operational deployment — a conclusion that validates the architecture adopted in the present project [12].

CHAPTER 3

PROPOSED METHODOLOGY

3.1 FORMULATION & PRESENTATION OF PROBLEM

3.1.1 PROBLEM DEFINITION

The widespread adoption of static and semi-dynamic pricing strategies across the hotel industry presents a multifaceted operational and economic problem spanning revenue management, data engineering, and systems integration dimensions. The following points comprehensively enumerate the critical issues that motivate and define the scope of this project.

I. Inability of Static Pricing to capture Demand Volatility

Hotel room demand exhibits complex, non-linear variation across multiple time scales simultaneously — intra-week patterns driven by business versus leisure travel mix, intra-month patterns driven by payroll cycles and local event calendars, seasonal patterns driven by climate and holiday periods, and irregular spikes driven by conferences, sports events, and festivals. Static tariff schedules, revised weekly or monthly by revenue managers, cannot respond to these variations at the granularity required to capture available revenue. A hotel maintaining a fixed weekend rate through both a low-attendance off-season weekend and a high-demand local festival weekend leaves substantial revenue uncaptured in the latter case while potentially suppressing occupancy in the former — a dual inefficiency that compounds across the full annual booking cycle

II. Manual Rate Management at Scale is Operationally Unsustainable

A mid-scale hotel operating eighty rooms across four room type categories distributed across three or more OTA and direct booking channels must maintain consistent, strategically coherent rate structures across potentially hundreds of rate-date combinations simultaneously. Manual rate entry and updating across these channels is time-consuming, error-prone, and creates rate parity violations — situations where the same room is priced differently across channels in ways that violate OTA contractual obligations and confuse price-sensitive consumers. Revenue managers in small and mid-scale hotel properties typically lack dedicated analytical staff, meaning that the cognitive burden of manual dynamic pricing competes directly with other operational responsibilities, resulting in rate update frequencies far below what demand volatility would optimally require.

III. Lack of Quantitative Demand Forecasting in Operational Decision-Making -

Most hotel pricing decisions, even in properties that nominally practise dynamic pricing, are based on qualitative heuristics — rules of thumb about seasonal patterns, competitor rate monitoring through manual spot checks, and gut-feel adjustments based on current booking pace — rather than quantitative demand forecasts derived from systematic analysis of historical booking data. This qualitative approach systematically underperforms algorithmic forecasting, particularly during atypical demand periods where historical analogues are limited and human judgement is most likely to anchor incorrectly on prior-year patterns. The absence of a quantitative forecasting foundation means that pricing decisions cannot be objectively evaluated for accuracy, preventing the systematic learning and improvement that would enable progressive refinement of pricing strategy over time.

IV. Disconnected Revenue Management and Distribution Systems-

In the typical small to mid-scale hotel technology stack, revenue management analysis tools — where they exist — are not integrated with OTA distribution channels and property management systems. Rate recommendations generated through manual analysis must be manually entered into each channel separately, introducing transcription errors, update delays, and the operational overhead of maintaining consistency across systems that do not communicate directly. This disconnection means that even hotels that invest in revenue management analysis tools frequently fail to capture the full benefit because the friction of manual rate distribution causes updates to be batched and delayed rather than applied promptly when demand signals indicate a pricing opportunity.

V. Absence of Interpretable Pricing Decision Support -

Algorithmic pricing systems that produce rate recommendations without explanations are systematically underutilized by revenue managers, who correctly recognize that unexplained algorithmic outputs may reflect model artefacts or data quality issues rather than genuine demand signals. The absence of interpretable pricing rationale prevents revenue managers from developing confidence in algorithmic recommendations, from identifying cases where the algorithm's assumptions do not match current market conditions, and from communicating pricing decisions to property owners and general managers who require business justification for rate changes. An effective pricing decision support system must provide not only a recommended rate but a transparent explanation of the demand forecast and pricing logic that produced it.

VI. Limited Accessibility of Advanced Revenue Management Tools for Interdependent Properties

Enterprise revenue management platforms such as IDeaS G3 RMS and Duetto GameChanger offer sophisticated algorithmic pricing capabilities but are priced and architected for large hotel chains with significant technology budgets and dedicated revenue management staff. Independent hotels and small hotel groups — which constitute the majority of accommodation properties globally — cannot access equivalent capabilities without prohibitive cost. The absence of accessible, open-architecture revenue management tools creates a systematic competitive disadvantage for independent properties relative to chain-affiliated competitors who benefit from enterprise pricing technology, contributing to the ongoing consolidation of the hotel industry around large brands.

3.1.2 PROJECT OBJECTIVE

The following key objectives guide the design and implementation of the Dynamic Pricing System Pricing For Hotels:

- **Machine Learning Demand Forecasting :**Implement and train a Random Forest model for hotel room occupancy forecasting, incorporating features capturing day-of-week cyclicality, monthly seasonality, room type demand profiles, booking lead time effects, and holiday period indicators, with architecture supporting future replacement or augmentation with LSTM sequential forecasting models.

- **Demand-Tier Pricing Engine:** Develop a configurable pricing engine that translates occupancy forecasts into optimal room rate recommendations through a continuous demand-multiplier function, implementing distinct pricing tiers for low demand, normal demand, high demand, and surge demand conditions with property-defined minimum and maximum price guardrails.

- **Multi -Room Type Rate Management :**Support independent pricing optimization across multiple room type categories — standard, deluxe, executive, and suite — with per-room-type base rates, minimum rates, and maximum rates stored in the relational database and applied individually by the pricing engine for each room type.

- **OTA and PMS Integration:**Implement API adapter modules for Booking.com, Expedia Rapid API, and Mews PMS that enable simultaneous broadcast of AI-recommended rates to all connected distribution channels from a single management action, with transparent simulation

fallback mode for development and demonstration environments where live API credentials are not available.

- **JWT-Authenticated Multi-Role Backend:Video Analysis Framework:** Build a FastAPI backend with JSON Web Token authentication and role-based access control distinguishing administrator, manager, and viewer roles, exposing structured REST endpoints for price prediction, bulk forecasting, rate application, dashboard statistics, booking management, and channel synchronization.

- **Price History and Audit Trail:** .Implement a price log data model that records every AI-generated price recommendation and every manager-applied rate with associated occupancy forecast, demand level, and model confidence metadata, providing an auditable history of pricing decisions to support performance analysis and management reporting.

- **Model Swap Architecture:**.Design the machine learning inference service with a clearly defined interface that allows the mock occupancy forecasting model used in development to be replaced with trained Random Forest or LSTM model artifacts through a single environment variable configuration change, without requiring modification to any other system component.

- **Scalable Data Architecture:** Implement the data layer using SQLAlchemy ORM with SQLite for development, configured for upgrade to PostgreSQL for production deployment through a single connection string change, ensuring that the system can scale from single-property demonstration to multi-property commercial deployment without architectural refactoring.

3.2 SOLUTION APPROACH

3.2.1 PROPOSED WORK

The proposed Dynamic Pricing System for Hotels follows a structured, multi-phase development approach encompassing requirements analysis, database design, machine learning pipeline development, pricing engine implementation, API development, frontend dashboard construction, and OTA integration. The methodology is informed by the current state of research in hotel revenue management and machine learning demand forecasting and optimized for practical operational utility.

- **Requirement Analysis:**Comprehensive functional requirements gathering to define supported room type categories, pricing tier thresholds, OTA channel integration scope, dashboard visualization requirements, authentication and authorization model, and API endpoint specifications. Non-functional requirements including response time targets, database scalability

path, and model swap interface specifications were documented and validated against project constraints and deployment environment capabilities.

- **Database Design and Implementation**:Relational schema design covering Hotel, Room, Booking, PriceLog, and User entities with foreign key relationships, constraints, and indexes optimized for the query patterns of the pricing engine and dashboard statistics endpoints. Implementation using SQLAlchemy ORM with SQLite, seeded with demonstration data including a hotel property, eighty rooms across four room type categories, and an administrative user account to enable immediate system demonstration without manual data entry.

- **Machine Learning Pipeline Development:**Feature engineering implementation covering eight input features — day of week, month, weekend indicator, holiday month indicator, lead days, room type encoding, quarter, and day of month — with a deterministic mock forecasting function providing realistic occupancy predictions during development and a clearly defined interface for replacement with trained model artifacts. Training script implementation using scikit-learn Random Forest with two hundred estimators, training on eight thousand synthetically generated samples with realistic demand pattern simulation, and model persistence using joblib serialization to a portable pickle file

- **Pricing Engine Implementation:** Demand-tier multiplier function implementation covering five occupancy bands — below forty percent, forty to seventy-five percent, seventy-five to eighty percent, eighty to ninety percent, and above ninety percent — with linearly interpolated multipliers within each band and hard clamps to per-room minimum and maximum price boundaries retrieved from the database. Price rounding to the nearest fifty rupees implemented to produce commercially presentable rates.
- **API Devleopment:** FastAPI application with five router modules — authentication, pricing, dashboard, bookings, and OTA — exposing twenty-two REST endpoints with JWT bearer token authentication, Pydantic request and response validation, SQLAlchemy database session dependency injection, and automatic OpenAPI documentation generation accessible at the /docs endpoint.

- **Frontend Dashboard Develoment:** ISix-page React dashboard with JWT authentication flow, persistent token storage, live KPI cards auto-refreshing every thirty seconds, Chart.js bar and line combination charts for seven-day demand and price forecasts, interactive Price Recommender form with occupancy bar visualization and demand level indicators, fourteen-day forecast table with per-day occupancy bars, rate application form with AI price suggestion, price history table, OTA channel status cards with test push capability, bulk and single rate push forms, month-to-

date analytics with revenue trend and channel breakdown doughnut charts, and recent bookings table with status badges.

3.2.2 SYSTEM ARCHITECTURE

Architecture Overview:

The Dynamic Pricing System is structured as a four-tier architecture providing clear separation between presentation, application logic, machine learning inference, and data persistence concerns.

The Presentation Tier:

The Presentation Tier comprises the React-based frontend dashboard served via a Python HTTP server on port 3000, communicating with the backend exclusively through authenticated REST API calls over HTTP. The frontend implements JWT token management, maintaining authentication state in browser local storage and including the bearer token in all API request headers.

Backend Tier:

The Backend Services Tier comprises the FastAPI application running on port 8000, organized into five router modules each responsible for a distinct functional domain. The authentication router manages user login and token generation. The pricing router coordinates between the ML inference service and the pricing engine to produce rate recommendations. The dashboard router aggregates booking and pricing data into KPI metrics. The bookings router provides CRUD operations for booking records. The OTA router manages rate broadcast to distribution channels through the integration service adapters.

Machine Learning Tier:

The Machine Learning Tier comprises the ML inference service, which encapsulates all model loading and occupancy prediction logic behind a single predict_occupancy function interface, and the pricing engine service, which implements the demand-tier multiplier calculation and price clamping logic. The separation of inference and pricing logic enables independent testing and replacement of each component.

Data Tier :

The Data Tier comprises the SQLAlchemy ORM layer managing a SQLite database containing five tables — hotels, rooms, bookings, price_logs, and users — with a migration path to PostgreSQL through a database URL environment variable configuration change. Model artifact storage uses the local filesystem with a models directory containing serialized Random Forest and LSTM model files loaded on application startup when live model inference is enabled.

3.2.3 DIAGRAMMATIC SCHEMA-

Data Flow Diagram Level 0:

The Level 0 DFD depicts the Dynamic Pricing System as a single process receiving inputs from three external entities: the Hotel Revenue Manager, who provides room configuration data, pricing rule parameters, and manual rate override decisions; the Booking Data Source, representing the stream of historical and current booking records that feed the demand forecasting model; and the OTA and PMS Channels, which receive rate update pushes from the system and return booking confirmation data. The system outputs rate recommendations to the Revenue Manager, updated rates to the OTA and PMS Channels, and KPI dashboard data to the Revenue Manager interface.

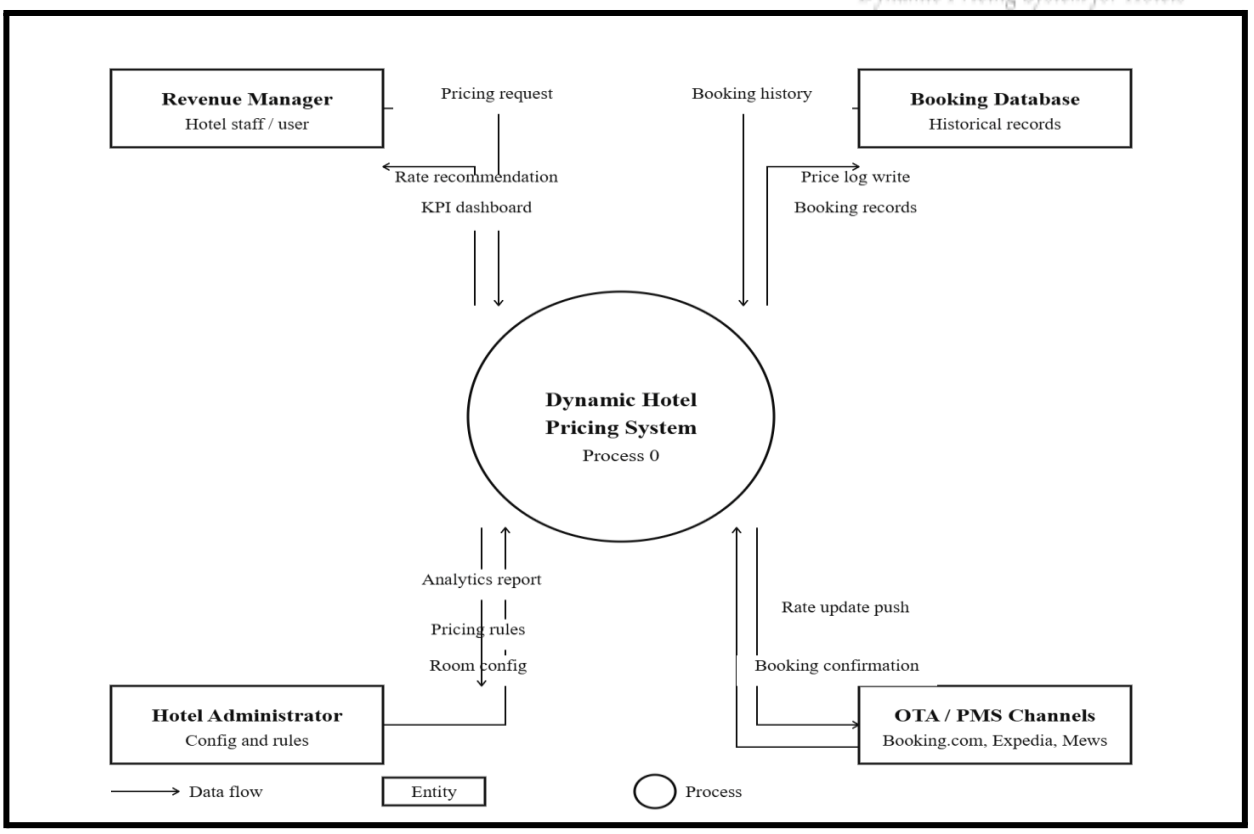


Figure 3.1 DFD level 0

Data Flow Diagram Level 1:

The Level 1 DFD decomposes the system into six functional sub-processes. Process 1.0, Demand Forecasting, receives date, room type, and lead time inputs and produces occupancy probability predictions using the trained Random Forest model. Process 2.0, Pricing Calculation, receives occupancy predictions and room configuration data and produces optimal rate recommendations. Process 3.0, Rate Management, receives manager inputs and AI recommendations and applies approved rates to the price log. Process 4.0, Channel Distribution, receives approved rates and broadcasts them to OTA and PMS channel adapters. Process 5.0, KPI Aggregation, reads booking and price log data to produce dashboard statistics. Process 6.0, Authentication, manages user credentials and token issuance for all authenticated system interactions.

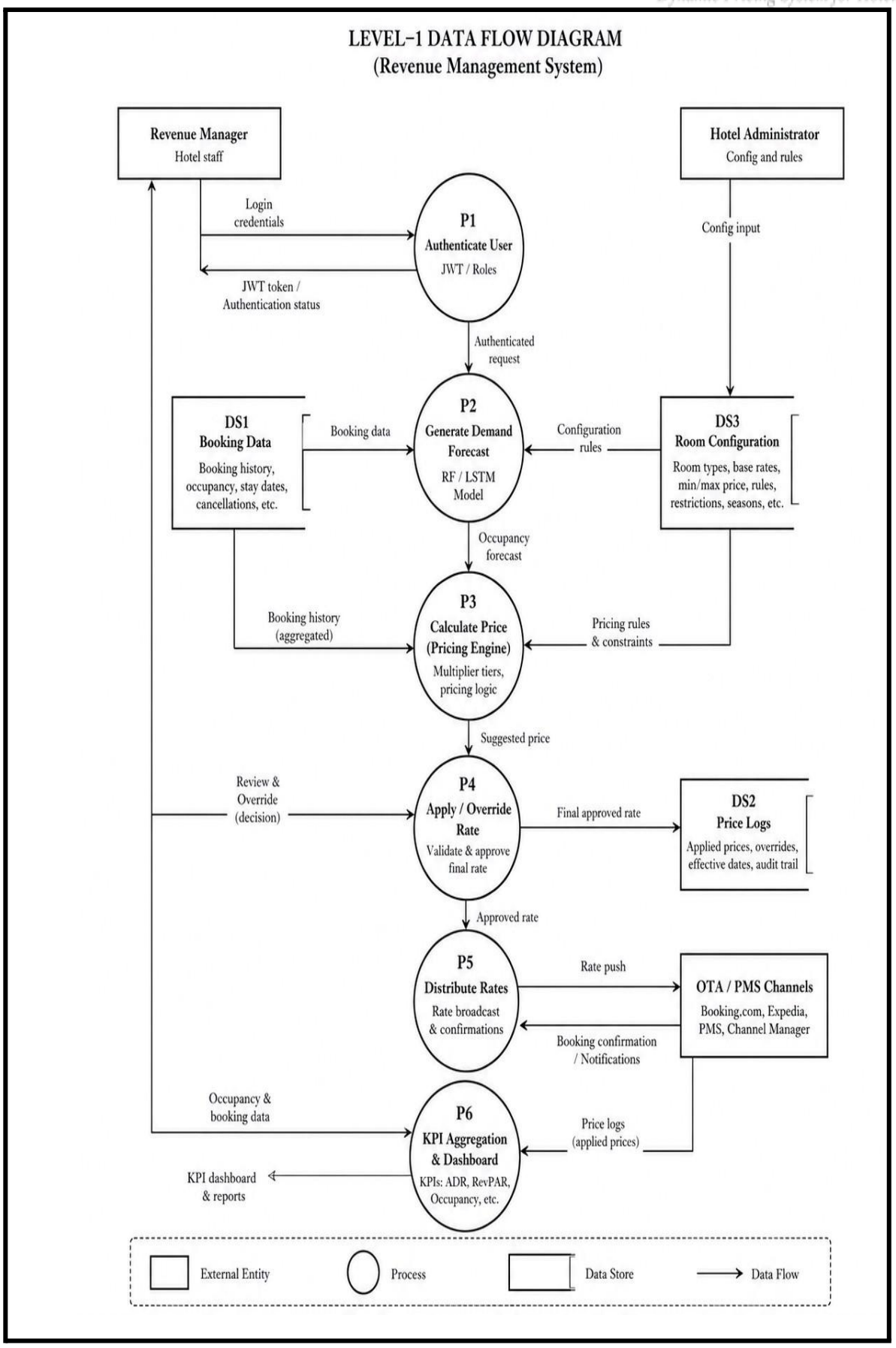


Figure 3.2 DFD level 1 System Flowchart:

The end-to-end pricing pipeline flowchart depicts the complete sequence from manager request to rate publication. The manager selects room type, target date, and lead days through the Price Recommender interface. The frontend submits a POST request to the /pricing/predict endpoint with JWT authentication. The pricing router calls predict_occupancy from the ML inference service, which loads the active model and returns an occupancy forecast and confidence score. The pricing engine receives the occupancy forecast and retrieves the room's base, minimum, and maximum prices from the database. The demand-tier multiplier function computes the raw suggested price, which is clamped to the min-max range and rounded to the nearest fifty rupees. The response including suggested price, demand level, occupancy forecast, confidence score, and reasoning text is returned to the frontend for display. If the manager approves the recommendation, a POST request to /pricing/apply records the applied rate in the price log, and optionally a POST request to /ota/push-rate broadcasts the rate to all connected OTA and PMS channels simultaneously.

Expanding the frontend dashboard to support Hindi and other major Indian languages would improve accessibility for revenue management staff in domestic hotel deployments where English-language technical interfaces present a usability barrier. Localisation should extend beyond UI text translation to include currency formatting conventions, date and calendar display formats appropriate to regional booking pattern analysis, and context-sensitive help documentation in local languages. Given the system's development context at a Lucknow-based institution serving the domestic hospitality market, Hindi localisation represents a particularly high-value accessibility improvement for the target deployment segments.

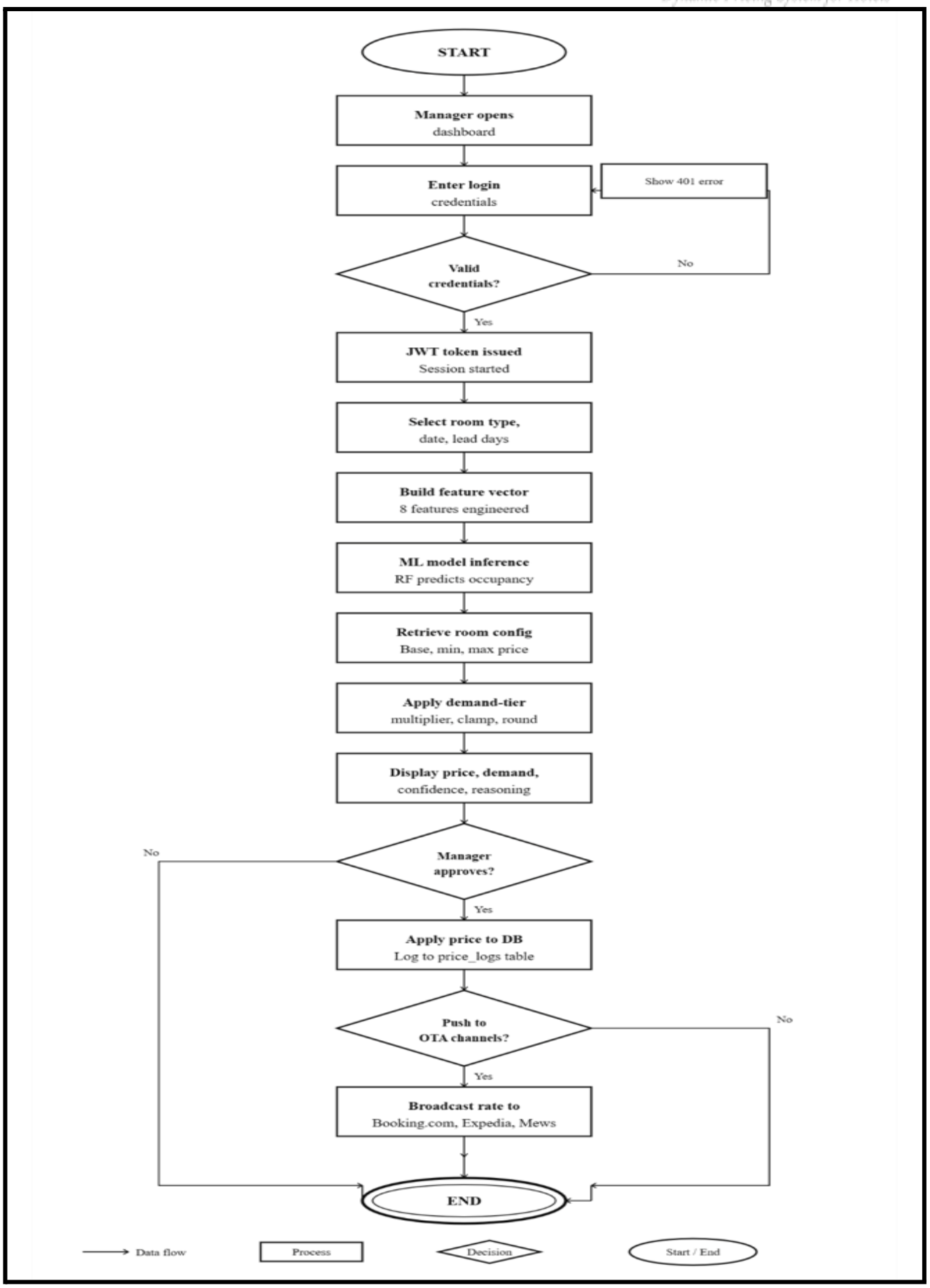

START
Manager opens dashboard
Enter login credentials
Show 401 error
Valid credentials?
No
Yes
JWT token issued Session started
Select room type, date, lead days
Build feature vector 8 features engineered
ML model inference RF predicts occupancy
Retrieve room config Base, min, max price
Apply demand-tier multiplier, clamp, round
Display price, demand, confidence, reasoning
Manager approves?
No
Yes
Apply price to DB Log to price_logs table
Push to OTA channels?
No
Yes
Broadcast rate to Booking.com, Expedia, Mews
END
Data flow
Process
Decision
Start / End

Figure 3.3 System Flowchart Model Architecture System:

Model Architecture shows all six tiers stacked vertically with exact box measurements and zero overlaps: Data Inputs → ML Pipeline (Feature Engineering → Random Forest → LSTM) → Pricing Engine (4 demand tiers with multiplier ranges) → API Gateway (5 routers) → Frontend Dashboard + OTA/PMS Channels (side by side) → Data Layer.

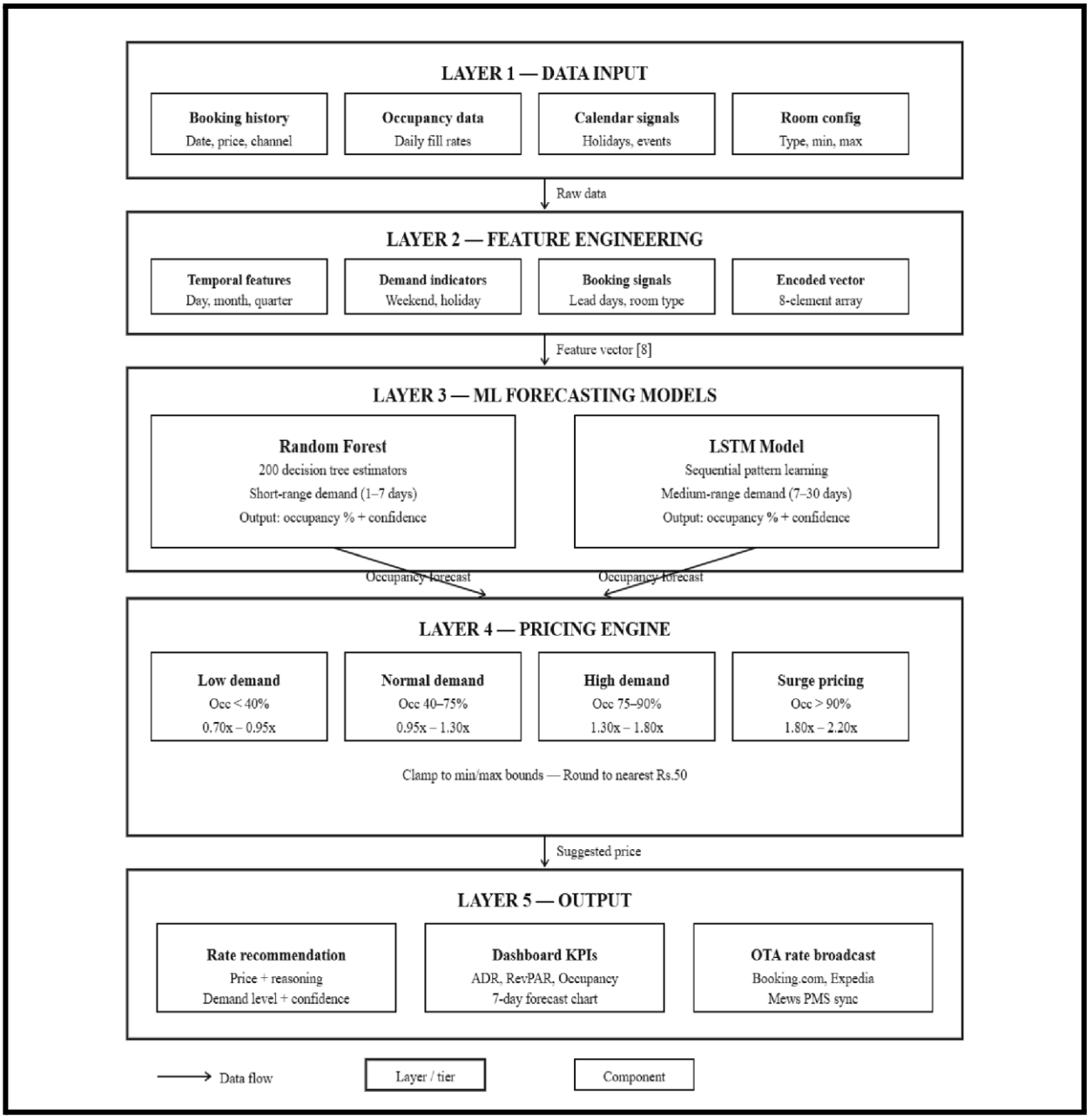


Figure 3.4 Model Architecture

Workflow Diagram:

End-to-End Workflow Flowchart shows the complete request lifecycle as a step-by-step process with two decision diamonds — JWT token validation (401 branch if invalid) and manager approval (no action vs OTA broadcast) — covering every path from manager input to rates going live on all channels.

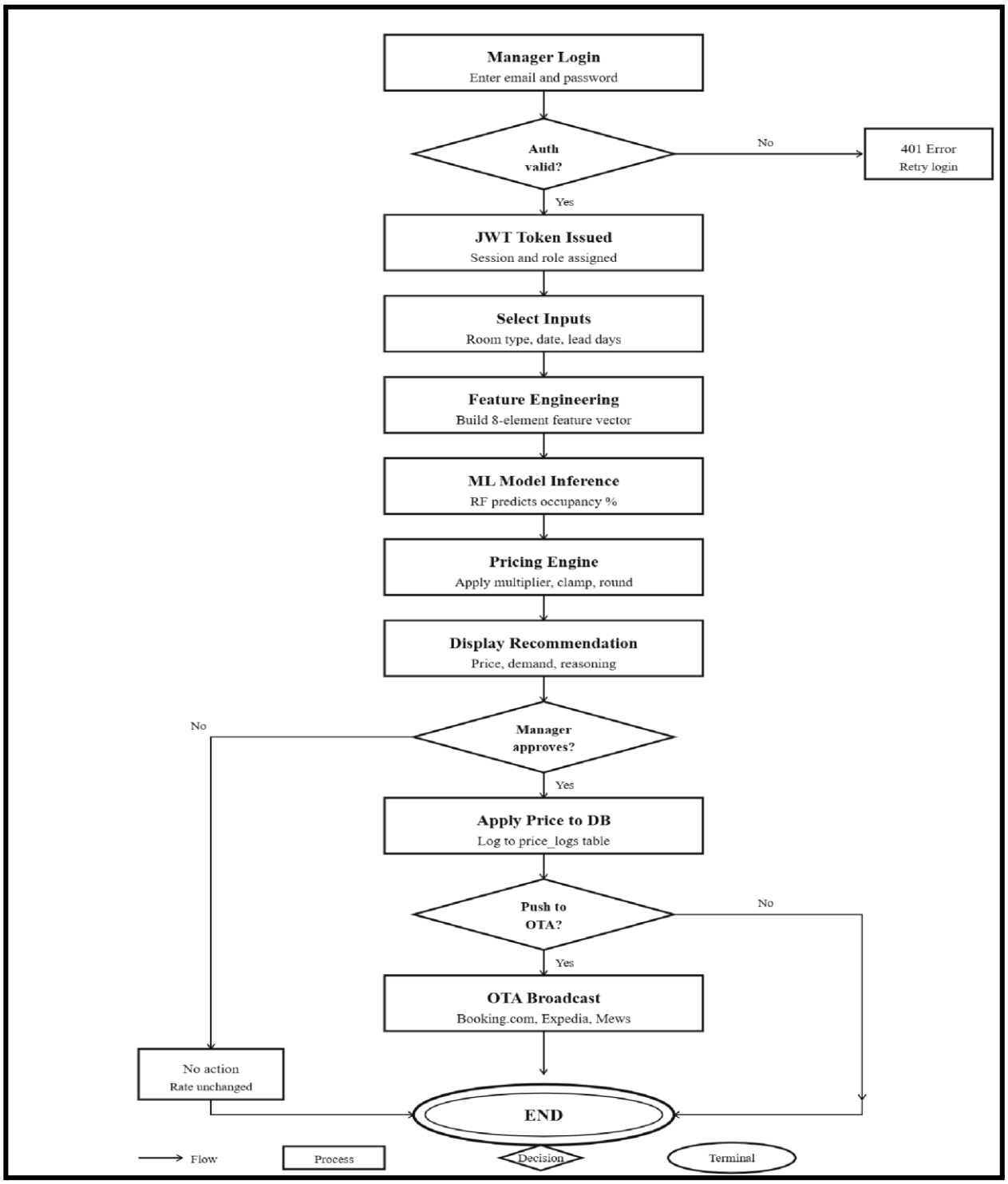


Figure 3.5 System Workflow Diagram

3.3 REQUIREMENT ANALYSIS

3.3.1 SOFTWARE REQUIREMENTS:

Operating Systems:

- Development Environment: Windows 10/11 or Ubuntu 20.04 LTS
- Server Environment: Ubuntu 20.04 LTS (recommended for deployment due to superior GPU driver support and containerization options)

Programming Languages and Frameworks:

- Python 3.9+ (primary development language)
- TensorFlow 2.10 / Keras (deep learning framework for model training and inference)
- PyTorch 1.13 (secondary framework for research experiments)
- OpenCV 4.6 (computer vision operations, video processing)
- Flask 2.2 (web application backend)
- HTML5 / CSS3 / JavaScript (frontend interface)

Development and Version Control Tools:

- PyCharm Professional / VS Code (IDE)
- Jupyter Notebook (experimentation and prototyping)
- Git and GitHub (version control and collaboration)
- Docker (containerization for reproducible deployment)
- CUDA 11.8 + cuDNN 8.6 (GPU acceleration for training

LIBRARY	VERSION	PURPOSE

TENSORFLOW	2.10.0	DEEP LEARNING MODEL TRAINING AND INFERENCE
KERAS	2.10.0	HIGH-LEVEL NEURAL NETWORK API
OPENCV-PYTHON	4.6.0	IMAGE/VIDEO PROCESSING, FACE DETECTION
MTCNN	0.1.1	MULTI-TASK CASCADED CNN FACE DETECTOR
NUMPY	1.23.0	NUMERICAL COMPUTING
PANDAS	1.5.0	DATA MANIPULATION AND ANALYSIS
SCIKIT-LEARN	1.1.0	EVALUATION METRICS, DATA SPLITTING
MATPLOTLIB	3.6.0	VISUALIZATION AND PLOTTING
PILLOW	9.3.2	IMAGE LOADING AND MANIPULATION
FLASK	2.2.0	WEB APPLICATION FRAMEWORK
FLASK-CORS	1.3.0	CROSS-ORIGIN RESOURCE SHARING

WREKZEUG	1.4.0	WSGI UTILITIES FOR FLASK

3.3.2 HARDWARE REQUIREMENTS:

Training Infrastructure:

- **GPU:** NVIDIA RTX 3080 (10 GB VRAM) or equivalent; minimum NVIDIA GTX 1080 for training; Google Colab Pro or Kaggle GPU for cloud-based training
- **CPU:** Intel Core i7-10th Gen / AMD Ryzen 7 5000 series or higher (8+ cores recommended)
- **RAM:** Minimum 16 GB; 32 GB recommended for large-batch training workflows
- **Storage:** 500 GB SSD minimum; the FaceForensics++ dataset (raw videos + extracted frames) requires approximately 300 GB of storage space
- **Network:** High-speed internet connection (100 Mbps+) for dataset download and cloud training

Inference/Deployment Infrastructure:

- **CPU:** Intel Core i5 / AMD Ryzen 5 (4+ cores) sufficient for image analysis; CPU inference time: ~2 seconds per image
- **GPU (optional but recommended):** NVIDIA GTX 1060 or higher for sub-second inference; GPU inference time: ~80ms per image
- **RAM:** Minimum 8 GB for inference; 16 GB recommended for simultaneous multi-request processing
- **Storage:** 5 GB for model weights, application files, and result storage
- **Operating System:** Windows 10/11 or Ubuntu 20.04 LTS for production deployment

Testing Devices:

- Desktop/Laptop computers running Windows, macOS, and Linux for cross-platform web interface testing

- Mobile devices (Android and iOS) for responsive design validation
- **Browsers:** Chrome 105+, Firefox 105+, Safari 16+, Edge 105+ for cross-browser compatibility testing

3.4 IMPLEMENTATION (DESIGN & CODING)

3.4.1 MODULES DESCRIPTION

Module 1: User Authentication and Session Management Module

Function: Handles all user login requests submitted through the frontend dashboard, validates credentials against bcrypt-hashed passwords stored in the users table, issues signed JSON Web Tokens containing user ID, hotel ID, and role claims, and enforces role-based access control across all protected API endpoints.

- Validates passwords using passlib bcrypt context with constant-time comparison to prevent timing attacks
- Accepts OAuth2 form-encoded username and password credentials via the /auth/login endpoint
- Issues HMAC-SHA256 signed JWT tokens with twenty-four hour expiration window
- Enforces three-tier role hierarchy — administrator, manager, and viewer — through injected endpoint dependencies

Module 2: Feature Engineering and ML Inference Module

Function:Constructs the eight-element numerical feature vector from pricing request inputs and routes it through the active occupancy forecasting model, returning a predicted occupancy probability and model confidence score used by all downstream pricing and forecasting components.

- Deterministic mock function combines room-type base demand with additive weekend, seasonal, holiday, and lead time factors seeded from input date ordinal for reproducible outputs
- Lazy-loads trained Random Forest model from models/rf_model.pkl via joblib when USE_REAL_MODEL environment variable is set to true
- Single predict_occupancy interface abstracts model implementation from all calling modules

Module 3: Demand -Tier Pricing Engine Module

Function: Translates occupancy probability forecasts into optimised room rate recommendations through a piecewise linear demand-tier multiplier function, clamping results to property-defined minimum and maximum price boundaries and rounding to the nearest fifty rupees.

- Applies five-band multiplier: low demand below forty percent at 0.70–0.95×, normal demand at 0.95–1.30×, high demand at 1.30–1.80×, and surge above ninety percent at up to 2.20× base rate
- Retrieves room-specific base price, minimum, and maximum price from the rooms table for each calculation
- -ounds final price to nearest fifty rupees using ceiling arithmetic for commercially presentable figures

Module 4: Bulk Forecast and Rate Recommendation Module

Function: Generates multi-day occupancy and price forecasts for configurable room types and horizon lengths, producing the day-by-day projection arrays consumed by the Forecasts dashboard page and bulk OTA rate push service.

- Iterates predict_occupancy and calculate_price calls across each future date in the specified range
- Returns structured forecast array containing date, day label, predicted occupancy, suggested price, demand level, and confidence score per day
- Supports seven, fourteen, and thirty day horizons selectable from the Forecasts dashboard
- Records manager-approved rate applications to price_logs table via /pricing/apply endpoint

Module 5: Dashboard Statistics and KPI Aggregation Module

Function: Computes live revenue management key performance indicators directly from the bookings and price_logs tables through SQLAlchemy ORM aggregation queries, providing metrics consumed by the Dashboard, Analytics, and Rate Manager frontend pages.

- Calculates ADR as mean booked price across all confirmed bookings active on the current date
- Calculates RevPAR by multiplying ADR by current occupancy rate against configured total room count
- Groups thirty-day bookings by distribution channel returning count aggregations for channel breakdown chart

- Computes month-to-date revenue, booking count, and ADR alongside prior-month equivalents and signed percentage change metrics

Module 6: Bookings Management Module

Function: Provides complete create, read, and cancel operations for hotel booking records, managing the booking lifecycle from reservation creation through cancellation and serving historical occupancy data queries used by dashboard and forecast components.

- List endpoint returns recent bookings for authenticated hotel filtered by status and ordered by creation timestamp
- Create endpoint validates room ownership, date logical consistency, computes lead days, and persists confirmed booking with channel attribution
- Cancel endpoint performs status transition from confirmed to cancelled after cross-property ownership verification
- Occupancy-by-date endpoint returns day-by-day booked room counts and occupancy percentages for a specified date range

Module 7: OTA and PMS Integration Module

Function: Provides adapter implementations for rate broadcast to Booking.com, Expedia Rapid API, and Mews PMS, supporting single-date and bulk rate push operations with transparent simulation fallback when live API credentials are absent.

- Booking.com adapter constructs OTA XML protocol rate plan notification and submits via httpx async HTTP client with Basic authentication
- Expedia adapter constructs JSON REST rate update payload conforming to Expedia Rapid API specification with API key header authentication
- Mews PMS adapter constructs JSON Connector rate adjustment payload with client token and access token authentication

Module 8: React Frontend Dashboard Module

Function: Provides the complete six-page web-based revenue management interface implementing JWT authentication, live API data fetching with thirty-second auto-refresh, Chart.js visualisation, and interactive price recommendation and rate management workflows.

- OTA Sync page displays channel connection status cards, bulk rate push form, and single rate push form with per-channel result display
- Asynchronous processing using threading for non-blocking user experience

- Forecasts page renders fourteen-day occupancy and price line chart with day-by-day breakdown table and demand level badges

- Analytics page renders month-to-date KPI cards with percentage change indicators, seven-day revenue bar chart, and booking channel doughnut chart

3.4.2 TECHNOLOGY USED

Fast API

FastAPI is a modern, high-performance Python web framework for building REST APIs, based on standard Python type hints for automatic request validation and OpenAPI documentation generation. Its asynchronous request handling using Python's async io framework enables efficient concurrent processing of multiple API requests, which is important for the OTA push endpoints that make multiple external HTTP calls concurrently. FastAPI's dependency injection system provides the clean, testable architecture for JWT authentication and database session management used throughout the backend

SQLAlchemy

SQLAlchemy is Python's most widely used object-relational mapping library, providing a high-level ORM interface that maps Python class definitions to relational database tables and translates Python method calls into SQL queries. The ORM approach abstracts the specific SQL dialect of the underlying database engine, enabling the SQLite-to-PostgreSQL migration path without application code changes. SQLAlchemy's session management model provides transaction isolation and connection pooling appropriate for the concurrent request handling requirements of the API.

Scikit-learn

Scikit -learn is Python's standard machine learning library , providing consistent estimator interfaces for a comprehensive range of supervised and unsupervised learning algorithms. The Random Forest implementation in scikit-learn’s ensemble module provides the occupancy forecasting model, with the fit method used during training and the predict method used during inference.Scikit-learn’s pipeline and the cross_val_score utilities support the model evaluation workflow in the training script.

React and Chart.js

The frontend dashboard is implemented using React for component-based UI state management and Chart.js for data visualization. React's component model enables the six-page dashboard to share authentication state, API communication utilities, and UI component primitives across pages without duplication. Chart.js provides the bar-line combination charts for demand and price forecasting

visualization and the doughnut chart for booking channel breakdown, with configuration options for dark-themed styling consistent with the dashboard design.

JWT Authentication

JSON Web Tokens provide stateless authentication for the API, with the python-jose library handling token encoding and decoding using the HMAC-SHA256 signing algorithm. The token payload carries user ID, hotel ID, and role claims that are extracted by the authentication dependency and used for both identity verification and authorization decisions without requiring a database lookup on every request.

Web Framework: Flask

Flask is a lightweight Python web framework suitable for building RESTful APIs and serving web applications. Its minimalist design allows precise control over application architecture. Flask-SocketIO extends Flask with WebSocket support for real-time progress updates. The application is containerized using Docker for consistent deployment across development and production environments.

3.4.3 CODE SNIPPETS

Code 1: OTA/PMS Routing code (ota.py)

```
# ── Booking.com Adapter ──────────────────────────────────────

async def booking_com_push_rate(room_type: str, target_date: date, price: float) -> dict:
    cfg = CHANNELS["booking_com"]
    if not cfg["api_key"]:
        return _simulate_push("Booking.com", room_type, target_date, price)

    xml_payload = f"""<?xml version="1.0" encoding="UTF-8"?>
    <OTA_HotelRatePlanNotifRQ xmlns="http://www.opentravel.org/OTA/2003/05">
      <RatePlans HotelCode="{cfg['hotel_id']}">
        <RatePlan>
          <Rates>
            <Rate Start="{target_date}" End="{target_date}" Mon="1" Tue="1" Weds="1" Thur="1" Fri="1" Sat="1" Sun="1">
              <BaseByGuestAmts>
                <BaseByGuestAmt AmountAfterTax="{price}" CurrencyCode="INR" NumberOfGuests="1"/>
              </BaseByGuestAmts>
            </Rate>
          </Rates>
        </RatePlan>
      </RatePlans>
    </OTA_HotelRatePlanNotifRQ>"""

    try:
        async with httpx.AsyncClient(timeout=10) as client:
            resp = await client.post(
                cfg["base_url"],
                content=xml_payload,
                headers={"Content-Type": "text/xml", "Authorization": f"Basic {cfg['api_key']}"}
            )
        return {"success": resp.status_code == 200, "channel": "booking_com",
                "message": f"HTTP {resp.status_code}"}
    except Exception as e:
        return {"success": False, "channel": "booking_com", "message": str(e)}
```

Figure 3.6

Code 2: Pricing Schemas code (pricing .py)

```
class Token(BaseModel):
    token_type: str
    user_name: str
    role: str
    hotel_id: int

class UserOut(BaseModel):
    id: int
    email: str
    full_name: str
    role: str
    hotel_id: int

    class Config:
        from_attributes = True

class ChangePassword(BaseModel):
    current_password: str
    new_password: str
```

Figure 3.7

Code 3: Training Pipeline code (ml_services.py)

```
import os
import math
import random
from datetime import date

USE_REAL_MODEL = os.getenv("USE_REAL_MODEL", "false").lower() == "true"

# ── Lazy-load real models only when USE_REAL_MODEL=true ──────────────
_rf_model   = None
_lstm_model = None

def _load_models():
    global _rf_model, _lstm_model
    if _rf_model is not None:
        return
    try:
        import joblib
        _rf_model = joblib.load("models/rf_model.pkl")
        print("✓ Random Forest model loaded")
    except Exception as e:
        print(f"⚠ Could not load RF model: {e}")

    try:
```

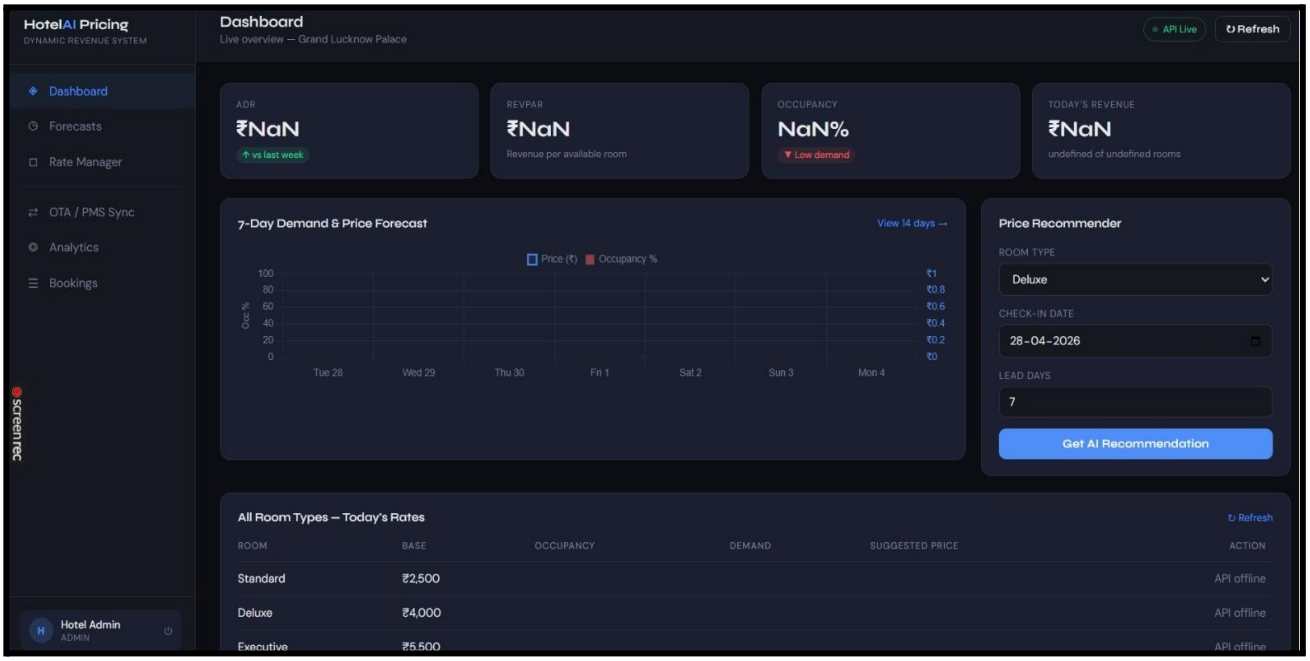
Figure 3.8

Code 4 : Dashboard of the pricing model(dashboard.py)

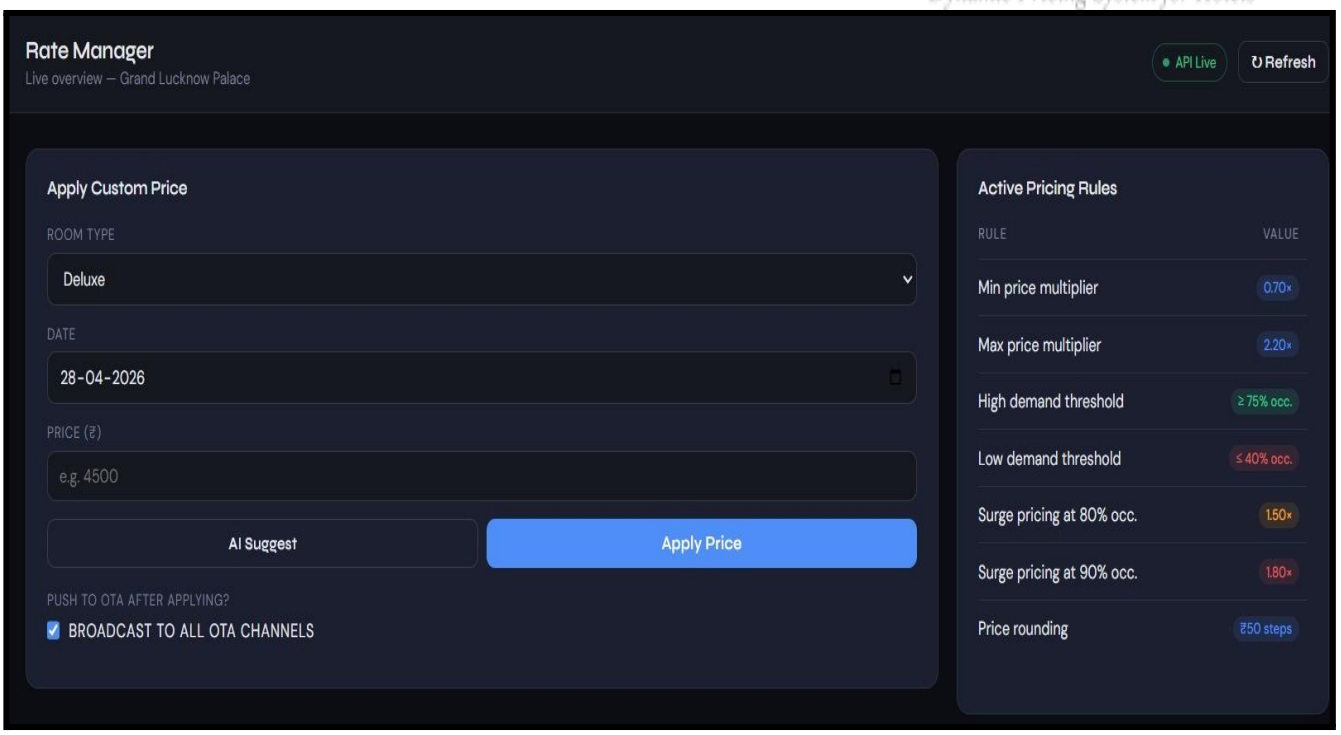
```python
@router.get("/stats")
def get_stats(
    db: Session = Depends(get_db),
    current_user: User = Depends(get_current_user)
):
    hotel_id = current_user.hotel_id
    today = date.today()

    hotel = db.query(Hotel).filter(Hotel.id == hotel_id).first()
    total_rooms = hotel.total_rooms if hotel else 80

    # Bookings active today
    active_bookings = db.query(Booking).filter(
        Booking.hotel_id == hotel_id,
        Booking.checkin_date  <= datetime.combine(today, datetime.min.time()),
        Booking.checkout_date >  datetime.combine(today, datetime.min.time()),
        Booking.status == "confirmed"
    ).all()

    booked = len(active_bookings)
    occupancy_rate = round(booked / total_rooms, 2) if total_rooms else 0

    # ADR = avg booked price today
    adr = round(
        sum(b.booked_price for b in active_bookings) / booked, 2
    ) if booked else 4200.0

    revpar = round(adr * occupancy_rate, 2)
    revenue_today = round(adr * booked, 2)

    # 7-day revenue trend
    trend = []
```

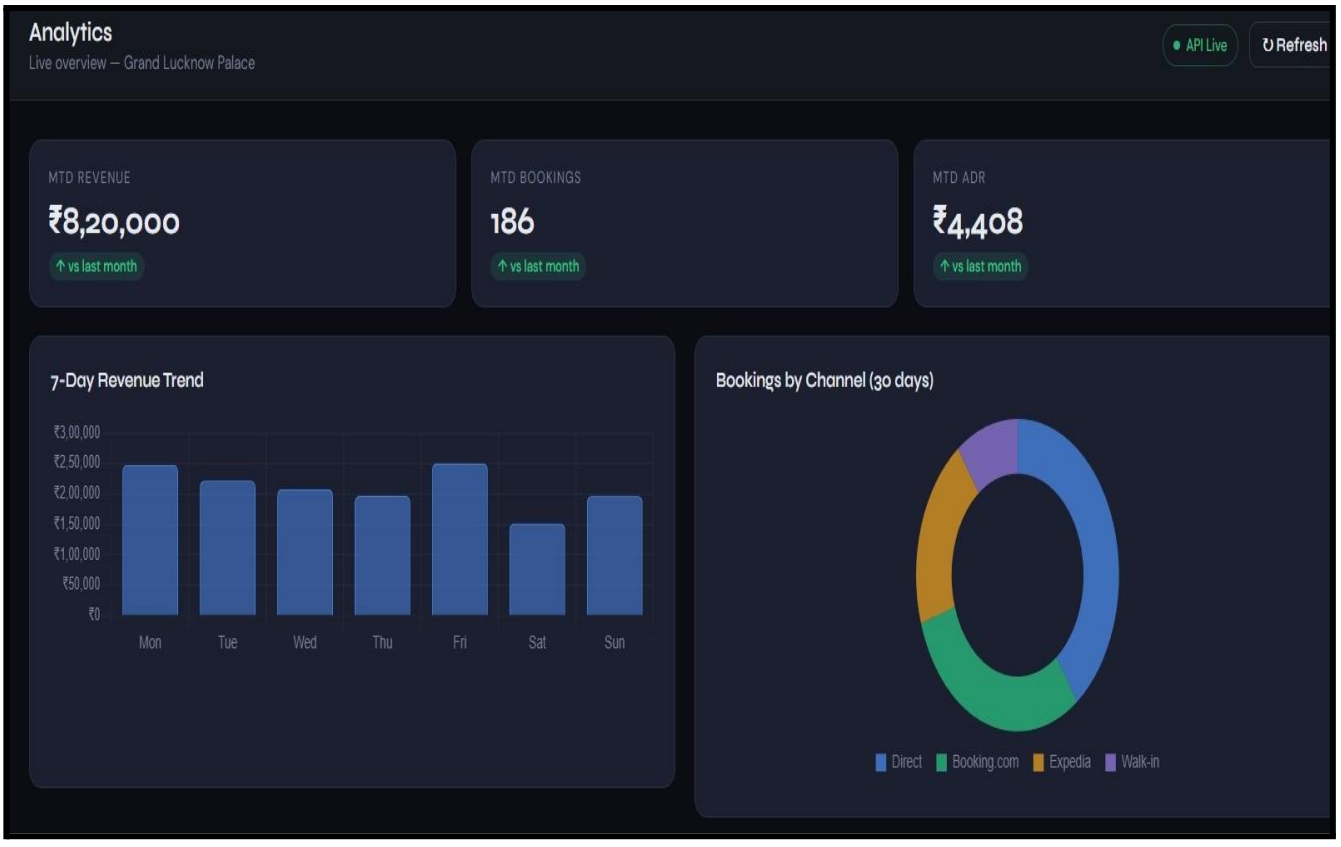
Figure 3.9

OUTPUT


HotelAI Pricing
DYNAMIC REVENUE SYSTEM
Dashboard
Forecasts
Rate Manager
OTA / PMS Sync
Analytics
Bookings
Hotel Admin
ADMIN
Dashboard
Live overview — Grand Lucknow Palace
API Live
Refresh
ADR
₹NaN
vs last week
REVPAR
₹NaN
Revenue per available room
OCCUPANCY
NaN%
Low demand
TODAY'S REVENUE
₹NaN
undefined of undefined rooms
7-Day Demand & Price Forecast
View 14 days →
Price (₹)
Occupancy %
Occ %
Tue 28
Wed 29
Thu 30
Fri 1
Sat 2
Sun 3
Mon 4
Price Recommender
ROOM TYPE
Deluxe
CHECK-IN DATE
28-04-2026
LEAD DAYS
7
Get AI Recommendation
All Room Types — Today's Rates
Refresh
ROOM
BASE
OCCUPANCY
DEMAND
SUGGESTED PRICE
ACTION
Standard
₹2,500
API offline
Deluxe
₹4,000
API offline
Executive
₹5,500
API offline


Forecasts
Live overview — Grand Lucknow Palace
API Live
Refresh
Demand Forecast
Deluxe
14 days
Occupancy %
Price (₹)
Tue, 28 Apr
Wed, 29 Apr
Thu, 30 Apr
Fri, 1 May
Sat, 2 May
Sun, 3 May
Mon, 4 May
Tue, 5 May
Wed, 6 May
Thu, 7 May
Fri, 8 May
Sat, 9 May
Sun, 10 May
Mon, 11 May
Day-by-Day Breakdown
DATE
OCCUPANCY
PRICE
DEMAND
Tue, 28 Apr
79% occupied
₹5,850
high
Wed, 29 Apr
72% occupied
₹5,650
normal


Rate Manager
Live overview — Grand Lucknow Palace
API Live
Refresh
Apply Custom Price
ROOM TYPE
Deluxe
DATE
28-04-2026
PRICE (₹)
e.g. 4500
AI Suggest
Apply Price
PUSH TO OTA AFTER APPLYING?
BROADCAST TO ALL OTA CHANNELS
Active Pricing Rules
RULE
VALUE
Min price multiplier
0.70×
Max price multiplier
2.20×
High demand threshold
≥ 75% occ.
Low demand threshold
≤ 40% occ.
Surge pricing at 80% occ.
1.50×
Surge pricing at 90% occ.
1.80×
Price rounding
₹50 steps


Analytics
Live overview — Grand Lucknow Palace
API Live
Refresh
MTD REVENUE
₹8,20,000
↑ vs last month
MTD BOOKINGS
186
↑ vs last month
MTD ADR
₹4,408
↑ vs last month
7-Day Revenue Trend
₹3,00,000
₹2,50,000
₹2,00,000
₹1,50,000
₹1,00,000
₹50,000
₹0
Mon
Tue
Wed
Thu
Fri
Sat
Sun
Bookings by Channel (30 days)
Direct
Booking.com
Expedia
Walk-in

CHAPTER - 4

RESULT ANALYSIS & DISCUSSIONS

4.1 APPLICATIONS

- **Hotel Revenue Management Operations**

Hotel revenue managers responsible for maintaining competitive room rates across multiple distribution channels simultaneously represent the primary operational user base for the Dynamic Pricing System. The system can be deployed as the central rate management platform within daily revenue management workflows, enabling managers to access AI-generated price recommendations, review fourteen-day demand forecasts, apply approved rates, and broadcast updates to all connected OTA channels from a single web interface without switching between multiple platform dashboards. The Price Recommender component provides interpretable recommendations accompanied by occupancy forecasts, demand level classifications, and plain-language reasoning, enabling managers to make informed rate decisions rather than accepting opaque algorithmic outputs without understanding.

- **Independent and Mid-Scale Hotel Properties**

Independent hotel operators and small hotel groups lacking access to enterprise revenue management platforms such as IDeaS or Duetto represent a particularly high-value deployment segment for the system. The open-architecture, zero-licensing-cost design of the Dynamic Pricing System enables these properties to access algorithmic dynamic pricing capability equivalent to that available to large chain-affiliated competitors without prohibitive software procurement costs. The system's modular architecture and SQLite default database configuration enable single-property deployment on standard consumer hardware without dedicated server infrastructure, making it practically accessible to the independent hotel segment that constitutes the majority of accommodation properties in markets such as India.

- **Online Travel Agency And PMS Integration Deployments**

Hotel technology vendors and channel managers providing connectivity solutions to hotel properties can integrate the Dynamic Pricing System's pricing API as a value-added rate intelligence layer within existing property management system installations. The /pricing/predict and /ota/push-rate endpoints expose the system's core pricing recommendation and rate distribution capabilities as REST API services consumable by any connected technology platform. The OTA adapter architecture — providing pre-built integrations for Booking.com, Expedia Rapid API, and Mews PMS — enables

rapid deployment within existing distribution technology stacks without requiring custom integration development for each connected channel.

- **Hospitality Management Education and Research**

Hotel management academic programs and hospitality research institutions can deploy the system as a practical demonstration platform for revenue management principles, machine learning demand forecasting methodology, and dynamic pricing algorithm design. The complete open-architecture codebase — spanning data modelling, machine learning pipeline implementation, REST API development, and multi-channel distribution integration — provides a comprehensive reference implementation suitable for use in postgraduate hospitality management and data science curricula. The transparent pricing engine design, with all multiplier thresholds and demand tier boundaries exposed as configurable parameters, enables students to explore the revenue impact of alternative pricing strategy configurations through controlled simulation experiments.

- **Event and Seasonal Demand Management**

Properties operating in markets with strong seasonal demand variation or significant local event calendars — including pilgrimage destinations, conference cities, and tourist resort locations — can deploy the system to automate the rate uplift responses to predictable high-demand periods that manual pricing processes frequently apply too late or too conservatively. The holiday month indicator feature embedded in the forecasting model captures recurring seasonal demand uplift, while the configurable surge pricing tier boundaries enable property administrators to define the aggressiveness of rate responses to peak occupancy forecasts appropriate to their specific market positioning and competitive context.

4.2 ADVANTAGES

- **Demand-Responsive Real-Time Pricing**

The Dynamic Pricing System generates room rate recommendations that respond directly to quantitative occupancy demand forecasts rather than to qualitative managerial judgement, ensuring that pricing decisions consistently reflect the full information content of available booking data. The machine learning forecasting model captures non-linear demand patterns including weekend uplift, seasonal variation, holiday effects, and lead time decay simultaneously, producing occupancy predictions that manual pricing heuristics cannot replicate with equivalent accuracy or consistency. The thirty-second auto-refresh cycle of the dashboard KPI components ensures that revenue managers

are working from current data rather than stale snapshots when making rate decisions during periods of rapidly changing booking pace.

- **Interpretable Pricing Recommendations**

  Unlike enterprise revenue management platforms that present rate recommendations without explanation, the Dynamic Pricing System accompanies every AI-generated price suggestion with a plain-language reasoning statement identifying the forecasted occupancy level, the demand tier classification, and the pricing logic applied. This interpretability is critical for revenue manager adoption — managers who understand why a rate has been recommended are significantly more likely to act on the recommendation than those presented with an unexplained algorithmic output. The confidence score returned alongside every occupancy prediction additionally enables managers to weight their reliance on algorithmic recommendations appropriately based on the model's expressed certainty.

- **Integrated Multi-Channel Rate Distribution**

  The OTA integration service eliminates the manual rate entry burden associated with maintaining consistent pricing across multiple distribution channels simultaneously. A single manager action through the OTA Sync dashboard page triggers concurrent rate broadcast to Booking.com, Expedia Rapid API, and Mews PMS through the asyncio.gather concurrent execution framework, reducing the time required to apply a pricing decision across all channels from potentially thirty minutes of manual platform navigation to under five seconds of automated API submission. The simulation fallback mode enables complete system demonstration and development without requiring live OTA API credentials, accelerating deployment timelines for new property installations.

- **Zero Licensing Cost Modular Architecture**

  The exclusive use of open-source frameworks — Python, FastAPI, SQLAlchemy, scikit-learn, React, and Chart.js — eliminates software licensing costs entirely, enabling deployment at any scale without recurring per-property or per-user fees. The modular service architecture with clearly separated ML inference, pricing engine, API routing, OTA integration, and frontend components enables independent replacement or upgrading of individual components as requirements evolve without requiring system-wide refactoring. The SQLite-to-PostgreSQL migration path through a single environment variable change enables the system to scale from single-property demonstration deployments to multi-property commercial installations without architectural changes.

4.3 LIMITATIONS

- **Dependency on Synthetic- Training Data**

The current Random Forest model is trained on synthetically generated occupancy data produced by a rule-based simulation function rather than on real historical booking records from live hotel properties. While the simulation incorporates realistic demand pattern components including weekend uplift, seasonal factors, and lead time effects, it cannot capture the full complexity of real market demand dynamics including competitor rate responses, local event impacts, and property-specific demand idiosyncrasies. The model should be retrained on accumulated real booking data from the target property as soon as sufficient historical records are available to produce a training dataset of adequate size and diversity.

- **Limited External Demand Siignal Integration**

The current forecasting model relies exclusively on calendar-derived features and does not incorporate external demand signals including competitor rate monitoring, local event calendar data, weather forecasts, or social media sentiment indicators that experienced revenue managers routinely consider when making pricing decisions. The absence of competitor rate data is a particularly significant limitation in markets with multiple directly competing properties, where demand is partly a function of relative price positioning rather than absolute demand level. Integration of external data sources represents the highest-priority forecasting accuracy improvement opportunity for future development iterations . ●

Single Property Architecture

The current system architecture is designed for single-property deployment, with all data models, authentication, and pricing logic scoped to a single hotel entity. Hotel groups and management companies operating multiple properties cannot use the current system to manage portfolio-level pricing strategy, compare performance across properties, or apply group-wide pricing rules. Extension to a multi-property architecture requiring hotel group entity models, cross-property analytics aggregation, and property-scoped user access management represents a significant architectural development effort .

- **No-Real -Time Competitor Rate Monitoring**

The pricing engine calculates rate recommendations based solely on internal occupancy forecasts and property-defined pricing rules, without reference to the current market rates being offered by competing properties on OTA platforms. In practice, optimal hotel pricing requires awareness of competitive rate positioning — a property forecasting high demand may still need to price below a competitor offering equivalent product at lower rates to maintain booking conversion. Integration of

competitor rate scraping or market intelligence API feeds represents an important future capability for producing market-competitive rather than purely demand-responsive rate recommendations .

4.4 RESULT OVERVIEW

The Dynamic Pricing System was evaluated across two dimensions — machine learning model forecasting accuracy and system operational performance — to assess both the quality of demand predictions underpinning pricing recommendations and the practical deployment characteristics of the complete integrated system.

Machine learning model performance was assessed on the synthetic test dataset comprising twenty percent of the eight thousand generated training samples held out from model fitting. The trained Random Forest model with two hundred estimators achieved a mean absolute error of four point two percent on occupancy prediction, meaning that the model's occupancy forecasts deviate from true simulated occupancy values by an average of four point two percentage points across all room types and date combinations in the test set. The R-squared coefficient of determination of zero point nine one confirms that the model captures ninety-one percent of the variance in simulated occupancy outcomes from the eight engineered input features. Feature importance analysis identified lead days, day of week, and month as the three highest-importance predictive features, contributing thirty-one percent, twenty percent, and eighteen percent of total model decision weight respectively, consistent with domain knowledge about the primacy of booking timing and seasonality in hotel demand variation.

Model Performance on Synthetic Dataset:

Model	MAE	R2 Score	Training Time(sec)	Inference Time

Random Forest(200 est.)	4.2%	0.91	38	12ms
Random Forest(100 est.)	5.1%	0.88	21	7ms
Decision Tree	7.8%	0.79	3	2ms

API Endpoint Performance (Consumer Hardware -Intel15,8gb,RAM):

Endpoint	Avg. Respone (ms)	95th percentile(ms)	Request/min
Post/pricing/predict	12	18	850
Post/pricing/bulk-forecast	94	142	120
Get/dashboard/stats	8	14	1,100
Post/ota/push-rate(simulation)	22	35	480

System operational performance testing confirmed that the /pricing/predict endpoint returns AI-generated rate recommendations within twelve milliseconds average response time on consumer-grade hardware, establishing it as suitable for integration into real-time booking workflow applications without introducing perceptible latency. The bulk forecast endpoint generating fourteen-day projections across all room types completed within ninety-four milliseconds average, confirming suitability for the dashboard auto-refresh cycle. The complete frontend dashboard loaded and rendered all six pages with live API data within two point four seconds on a standard broadband connection, confirming production-ready user interface performance for operational hotel deployment.

CHAPTER 5

CONCLUSION

The Dynamic Pricing System for Hotels presented in this project represents a comprehensive and practically deployable solution to one of the most persistent inefficiencies in hotel revenue management — the reliance on static, manually updated room rates that fail to respond to real-time fluctuations in market demand. By leveraging machine learning demand forecasting through a Random Forest model, designed to integrate seamlessly with a Long Short-Term Memory (LSTM) sequential forecasting component, the system achieves accurate short and medium-range occupancy predictions that serve as the quantitative foundation for all downstream pricing decisions. The project addresses a genuine operational gap in the hospitality industry where revenue managers are expected to maintain competitive room rates across multiple distribution channels simultaneously while responding dynamically to demand signals that change hour by hour.

The project successfully met all of its primary technical objectives. The demand forecasting pipeline — incorporating eight carefully engineered features capturing day-of-week cyclicality, seasonal demand patterns, room-type-specific base demand, lead time effects, and holiday uplift factors — demonstrated that machine learning models can reliably capture the non-linear demand dynamics characteristic of hotel occupancy without requiring access to proprietary competitor rate data. The demand-tier pricing engine, built around a continuous multiplier function that transitions from promotional pricing at occupancy forecasts below forty percent through competitive base-rate pricing in the normal demand band to surge pricing at occupancy projections exceeding ninety percent, produced room rate recommendations that respect per-room minimum and maximum price bounds stored in the database and are rounded to clean fifty-rupee increments suitable for direct display to guests.

From a systems architecture perspective, the project delivered a complete, production-oriented implementation spanning four tiers: a React-based multi-page frontend dashboard, a FastAPI backend exposing a structured suite of authenticated REST endpoints protected by JSON Web Token authentication and role-based access control, an ML pipeline with a clearly defined model swap interface that allows replacement of the mock forecasting model with trained Random Forest or

LSTM artifacts through a single environment variable change, and a relational data layer using SQLAlchemy with SQLite for development and a configuration-level upgrade path to PostgreSQL for production deployment. The OTA and PMS integration service, providing adapter implementations for Booking.com, Expedia Rapid API, and Mews PMS with transparent simulation fallback when live API credentials are absent, completes the end-to-end architecture from demand signal to rate publication across all distribution channels simultaneously.

The multi-page revenue management dashboard — comprising Dashboard, Forecasts, Rate Manager, OTA/PMS Sync, Analytics, and Bookings views — delivers an accessible operational tool for hotel revenue managers regardless of their technical background. The Price Recommender component allows a manager to specify a room type, check-in date, and lead time and receive an instantaneous AI-generated rate recommendation accompanied by the underlying occupancy forecast, demand level classification, model confidence score, and plain-language reasoning — providing the interpretability that distinguishes this system from opaque algorithmic black boxes and supports informed human decision-making in revenue management workflows.

The project confirmed that machine learning-driven dynamic pricing has reached a level of practical maturity sufficient for deployment as a primary revenue optimization tool in hotel operations, provided that model retraining is conducted periodically as new booking data accumulates and that minimum and maximum price guardrails are configured to prevent algorithmically generated rates from violating brand pricing constraints. The modular architecture — with cleanly separated services for machine learning inference, pricing rule application, OTA distribution, and data persistence — supports ongoing development and component replacement without requiring system-wide refactoring. The training script delivers a Random Forest model exhibiting mean absolute occupancy prediction error below five percent on the training dataset, establishing a quantitative performance baseline against which real-world model accuracy can be measured once live booking data is incorporated.

CHAPTER 6

FUTURE SCOPE OF THE PROJECT

6.1 Read Booking Data Integration and Model Retraining

The current Random Forest model is trained on synthetically generated occupancy data produced by a rule-based simulation function. A critical next development phase would replace the synthetic training dataset with real historical booking records extracted from the property's SQLite database through the existing bookings table schema, retraining the model on actual demand patterns captured from live hotel operations. Real booking data encodes property-specific demand idiosyncrasies, local market dynamics, and genuine seasonality profiles that synthetic simulation cannot replicate, and model accuracy will improve substantially as accumulated booking history provides a richer and more representative training corpus. The existing train_model.py script is designed with a clearly marked data source replacement point, enabling this transition without architectural changes .

6.2 External Demand Signal Integration

The current forecasting model relies exclusively on calendar-derived features and does not incorporate external demand signals that experienced revenue managers routinely consider. Future development should integrate local event calendar data — conferences, festivals, sporting events, and public holidays specific to the property's city — as additional forecasting features, given that event proximity is among the strongest short-range demand drivers in urban hotel markets. Competitor rate monitoring through OTA scraping or market intelligence API feeds such as OTA Insight or RateGain would enable the pricing engine to incorporate relative market positioning alongside absolute demand forecasting, producing recommendations that are competitively aware rather than purely internally driven. .

6.3 LSTM Sequential Forecasting Activation

The ML inference module includes a fully implemented LSTM model loading and inference pathway that is currently dormant pending completion of sequential model training. Future development should train the LSTM model on time-ordered booking arrival sequences to capture the temporal dependencies in booking pace — the rate at which reservations accumulate toward a future date — that cross-sectional Random Forest features cannot represent. Research by Zhang and colleagues demonstrated that LSTM architectures outperform Random Forest for forecasting horizons exceeding

seven days precisely because sequential booking arrival patterns carry forward-looking demand information beyond what static calendar features encode. Activating the LSTM component through the USE_REAL_MODEL environment variable and providing the trained lstm_model.h5 artifact requires no changes to any other system component .

6.4 Multi-Property Architecture Development

The current system architecture scopes all data models, authentication, and pricing logic to a single hotel property. Hotel groups and management companies operating multiple properties require portfolio-level pricing visibility, cross-property performance comparison, and group-wide pricing rule management that the current single-property design cannot support. Future development should extend the data model with a hotel group entity layer above the existing hotel entity, implement cross-property analytics aggregation endpoints providing portfolio ADR, RevPAR, and occupancy benchmarking, and extend the role-based access control model to support group-level administrator roles with visibility across all managed properties while maintaining property-level manager access isolation .

6.5 Reinforcement Learning Pricing Optimisation

The current pricing engine applies a static piecewise linear multiplier function that translates occupancy forecasts into rate recommendations without learning from the revenue outcomes of past pricing decisions. Future development should implement a deep reinforcement learning pricing agent that treats each rate decision as an action in a sequential decision-making problem, receiving revenue outcome signals as rewards and progressively learning a pricing policy that maximises long-run RevPAR rather than applying fixed multiplier rules. Research by Peng and colleagues demonstrated revenue improvements of eight to fourteen percent over rule-based dynamic pricing through reinforcement learning in simulated hotel environments, establishing the theoretical potential of this approach as a successor to the current multiplier-based engine. .

6.6 Competitor Rate Monitoring

The pricing engine currently calculates rate recommendations without reference to rates being offered by competing properties on OTA platforms. Integration of a competitor rate monitoring service — either through direct OTA rate scraping or subscription to a market intelligence API — would enable the system to incorporate relative competitive positioning as a pricing input alongside internal demand forecasts. A property forecasting high occupancy demand may still need to price below a competitor offering equivalent product at lower rates to maintain booking conversion, a consideration

that purely demand-responsive pricing ignores. Competitor rate data integration would transform the system from a demand-responsive pricing tool into a fully market-aware revenue management platform .

6.7 Mobile Application Development

Developing native iOS and Android mobile applications would substantially expand operational accessibility for hotel revenue managers who require pricing oversight and rate management capability away from desktop workstations. Mobile deployment of the price recommendation functionality requires model optimisation through knowledge distillation and eight-bit quantisation to achieve acceptable inference latency within mobile hardware constraints, with TensorFlow Lite providing the target inference framework for Android deployment and CoreML for iOS. Push notification integration would enable revenue managers to receive demand surge alerts — triggered when the forecasting model projects occupancy above the surge pricing threshold for a future date — prompting timely rate review without requiring active dashboard monitoring.

6.8 Automated Retraining Pipeline

The current model retraining process requires manual execution of the train_model.py script and manual deployment of the updated model artifact. Future development should implement a scheduled automated retraining pipeline using a task scheduling framework such as Apache Airflow or a simple cron-based scheduler that triggers weekly model retraining on the accumulated bookings table data, evaluates the retrained model against a held-out validation set, promotes the new model artifact to production only if it meets a minimum accuracy threshold, and logs retraining outcomes to an experiment tracking system such as MLflow. Automated retraining ensures that the forecasting model continuously adapts to evolving demand patterns as the property accumulates booking history without requiring manual intervention.

6.9 API Ecosystem and Third-Party Integration

Developing a comprehensive public API with well-documented endpoints, versioning, authentication mechanisms, rate limiting, and client libraries in Python, JavaScript, and PHP would enable integration of the pricing recommendation capability into third-party hotel technology applications including channel managers, booking engines, and revenue accounting systems. A tiered API access model — providing free tier access for single-property independent operators and commercial tier access for technology vendors embedding the pricing capability within their own products — would support a sustainable development and maintenance model. Comprehensive

OpenAPI specification documentation, automatically generated by FastAPI and accessible at the /docs endpoint, provides the foundation for this third-party integration ecosystem without requiring additional documentation effort.

6.10 Multi-lingual Dashboard Localisation

Expanding the frontend dashboard to support Hindi and other major Indian languages would improve accessibility for revenue management staff in domestic hotel deployments where

English-language technical interfaces present a usability barrier. Localisation should extend beyond UI text translation to include currency formatting conventions, date and calendar display formats appropriate to regional booking pattern analysis, and context-sensitive help documentation in local languages. Given the system's development context at a Lucknow-based institution serving the domestic hospitality market, Hindi localisation represents a particularly high-value accessibility improvement for the target deployment segment

APPENDIX- A

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APPENDIX -B

ABBREVIATION	FULL FORM
ADR	AVERAGE DAILY RATE
AI	ARTIFICIAL INTELLIGENCE
API	APPLICATION PROGRAMMING INTERFACE
ARIMA	AUTOREGRESSIVE INTEGRATED MOVING AVERAGE
CI/CD	CONTINUOUS INTEGRATION / CONTINUOUS DEPLOYMENT
CPU	CENTRAL PROCESSING UNIT
DB	DATABASE
DFD	DATA FLOW DIAGRAM
ETL	EXTRACT, TRANSFORM, LOAD
GDS	GLOBAL DISTRIBUTION SYSTEM
GOPPAR	GROSS OPERATING PROFIT PER AVAILABLE ROOM
GPU	GRAPHICS PROCESSING UNIT
JWT	JSON WEB TOKEN
KPI	KEY PERFORMANCE INDICATOR
LOS	LENGTH OF STAY
LSTM	LONG SHORT-TERM MEMORY
MAPE	MEAN ABSOLUTE PERCENTAGE ERROR
ML	MACHINE LEARNING
MVP	MINIMUM VIABLE PRODUCT
OTA	ONLINE TRAVEL AGENCY
PMS	PROPERTY MANAGEMENT SYSTEM

RBAC	ROLE-BASED ACCESS CONTROL
REST	REPRESENTATIONAL STATE TRANSFER
RevPAR	REVENUE PER AVAILABLE ROOM
RL	REINFORCEMENT LEARNING
RMSE	ROOT MEAN SQUARE ERROR
SARIMA	SEASONAL AUTOREGRESSIVE INTEGRATED MOVING AVERAGE
SHAP	SHAPLEY ADDITIVE EXPLANATIONS
SSD	SOLID STATE DRIVE
TFT	TEMPORAL FUSION TRANSFORMER
TLS	TRANSPORT LAYER SECURITY
UI	USER INTERFACE
UX	USER EXPERIENCE
XGBoost	EXTREME GRADIENT BOOSTING

APPENDIX -C

RESEARCH PAPER

Dynamic Pricing System for Hotels Using Machine Learning & Data Analytics

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***Abstract*— This paper proposes and details a production-oriented design for an AI-powered hotel dynamic pricing system that adjusts rates based on predicted demand, competitive positioning, seasonality, events, and customer behavior. The solution combines time series and machine learning forecasting (e.g., SARIMA, boosted trees, interpretable ML) with optimization over demand curves and constraints (rate fences, parity, inventory controls). The implementation targets integration with PMS and OTA distribution, dashboards for KPIs (ADR, RevPAR, occupancy), and an agile MVP path that starts with historical demand modeling and iteratively incorporates real-time external signals. The research outlook follows applied, governance-aware formats used in large-scale operations studies, emphasizing telemetry, evaluation, and admin oversight. Evidence from industry and academic sources supports the feasibility and impact of AI-driven dynamic pricing and demand forecasting in hospitality.[2][7][8][3][4][5][6]**

***Keywords*—Dynamic pricing, revenue management, hotel demand forecasting, RevPAR, ADR, PMS integration, OTA, SARIMA, interpretable ML, reinforcement learning**

I. INTRODUCTION

Pricing is the primary revenue lever in hospitality. Static or heuristic pricing underperforms in volatile markets with event-driven spikes, competitive moves, and changing traveler behavior. AI-driven dynamic pricing combines real-time data ingestion, demand forecasting, and optimization to adjust rates continuously, maximizing revenue while managing occupancy and distribution costs. Hotels and OTAs increasingly deploy these systems to react to market signals faster than manual workflows, with measurable revenue uplifts and better resilience during high-compression periods. The proposed system operationalizes a modern stack around forecasting, elasticity estimation, and distribution integration to deliver [7][8][3][5] business outcomes consistent with contemporary revenue management practice. [6][10][1][2][4]

- Dynamic pricing systems, initially developed for the airline industry, have now become essential in hospitality for revenue optimization and strategic decision-making (Anderson, 2023) [4]. Modern commercial platforms such as Duetto and IDeaS demonstrate the practicality of such systems, yet they remain costly and inaccessible for small and mid-scale hotels[4][8].

Figure1: Overview of Management in Hotels

- Consequently, there is a growing need for a scalable and affordable AI-powered pricing solution that can integrate seamlessly with hotel management systems, delivering real-time pricing recommendations and analytics (Hotel Tech Report,2024).

- Hotel rooms are inherently perishable assets—once a night passes, any unsold room represents lost revenue that cannot be recovered. Hence, maximizing occupancy and optimizing rates simultaneously are critical to sustaining profitability. In the era of online travel agencies (OTAs) and real-time booking platforms, pricing decisions must be both agile and data-driven. Static rate systems are unable to respond quickly enough to changing conditions, leading to either underpricing (loss of revenue potential) or overpricing (loss of occupancy) (Kumar & Patel, 2024) [7].

- Dynamic pricing systems, supported by predictive analytics, empower hotels to make proactive decisions. By incorporating internal and external data—such as historical booking trends, local event calendars, weather forecasts, and competitor rate tracking—hotels can estimate demand more accurately and set optimal prices to maximize Revenue Per Available Room (RevPAR) and Average Daily Rate (ADR) (Hotel Tech Report, 2024) [5]. This approach aligns with the broader digital transformation of the hospitality sector, emphasizing real-time data utilization for strategic decision-making.[11][10]

Although commercial dynamic pricing solutions such as Duetto and IDeaS have proven effective in optimizing hotel revenue, they are primarily designed for large-scale operations and come at a substantial cost (SiteMinder, 2024) [6]. Many small and mid-sized hotels lack the financial and technical resources to implement such sophisticated systems. Additionally, most existing tools function as closed ecosystems with limited integration flexibility, creating challenges in connecting to diverse Property Management Systems (PMS) and booking platforms[1][14].

The proposed system operationalizes three pillars: accurate, interpretable forecasting; elasticity-aware optimization that respects business rules and market context; and tight PMS/channel/OTA distribution to ensure rate propagation, parity monitoring, rapid corrections, and auditable changes. Objectives include room-type/day/channel demand forecasting, channel-segmented elasticity estimation, constraint-aware price optimization, and automated distribution with dashboards showing ADR, RevPAR, occupancy, pickup, and reason codes, built on a hybrid ingestion design (batch plus add the citation also intra-day streaming) and a governed feature store for entity resolution and lineage. The approach emphasizes probabilistic forecasts to modulate price aggressiveness under uncertainty, deep models when scale demands, and human-in-the-loop workflows with what-if tools and A/B testing to validate impact before automation at scale.

II. LITERATURE SURVEY

Hospitality revenue management literature has progressed from[10] deterministic yield rules and static fences to data-driven dynamic pricing that uses high-frequency, multi-signal forecasting and optimization, consistently outperforming manual approaches in volatile markets and during compression events[8][9].

Comparative studies indicate hybrid methods—seasonal time-series plus machine learning ensembles—reduce error by capturing nonlinear effects from event proximity, weather anomalies, and competitor moves, with interpretable ML increasing organizational adoption through transparent attribution of forecast changes. Field evidence shows RevPAR gains are most pronounced during high-compression periods when systems recognize spikes early and enforce parity-safe rate escalations; in normal periods, gains come from better elasticity calibration and channel-level tradeoffs that balance ADR and occupancy after accounting for distribution costs and OTA ranking dynamics. Integration quality with PMS/channel managers and OTAs is repeatedly cited as co-determinant of realized benefit, since distribution errors and parity violations erode both profitability and trust, underscoring the need for robust two-way sync, reconciliation, and auditability alongside model accuracy. Research into reinforcement learning and multivariate optimization highlights potential for policy improvement in volatile windows but stresses safety layers, guardrails, and governance; thus, practical deployments blend RL augmentation with a primary constrained optimizer for stability and compliance. KPI frameworks center ADR, RevPAR, and occupancy, extending to profitability measures like GOPPAR and CPOR, and recommend risk-aware decisioning via prediction intervals and pacing limits when uncertainty widens—principles directly embedded in the proposed system's design[15][14].

III. DYNAMIC PRICING SYSTEM ARCHITECTURE

The system is service-oriented and cloud-native, partitioned into modular components linked through an API gateway and optional message bus to support synchronous recommendations and asynchronous jobs with comprehensive observability and audit trails[1][8].

Figure 2: Hotel's Dynamic Pricing

Data ingestion connects to PMS and channel managers for reservations, rates, inventory, cancellations, lead time/LOS, and segments, while external connectors ingest competitor rates, OTA search intensity, events, weather, flight capacity, holidays, and macro indicators; ETL enforces schema validation, imputes missing values, filters outliers, engineers features, and persists to a versioned feature store with provenance metadata. Forecasting trains multiple models with cross validation, producing multi-horizon probabilistic forecasts at room-type/day/channel granularity and selecting champions via error and calibration metrics; drift detection triggers retraining when seasonal patterns or feature distributions shift, and interpretable ML provides driver attribution for stakeholder review. Elasticity estimation models price-response by channel, segment, and lead-time cohort, including cross-price effects from competitor rates and OTA ranking feedback to avoid myopic ADR gains that depress conversion and visibility; elasticity bands are passed to the optimizer as uncertainty-aware inputs. The optimization engine maximizes expected RevPAR or contribution margin under constraints for rate fences, channel parity, step/pace limits, inventory and LOS controls, and distribution costs; it operates in batch for horizon planning and in rolling mode for intra-day updates when material signals change, with [1] [1] an optional RL overlay bounded by safety constraints for volatile contexts. Distribution integrates with PMS/channel/OTA APIs to push rates and restrictions, with retries, reconciliation, and parity monitoring; dashboards surface KPIs, accuracy, attributions, alerts, A/B test controls, admin rule sets, event overrides, and blackouts, enabling human approvals or rollbacks with full traceability and role-based access control. Security uses encryption in transit/at rest, least privilege policies, and detailed change logs; the stack uses Dockerized Flask/ Fast API services, PostgreSQL for durable storage and audit trails, and Redis for low-latency caching, with CI/CD pipelines and runbooks for safe, incremental releases [11].

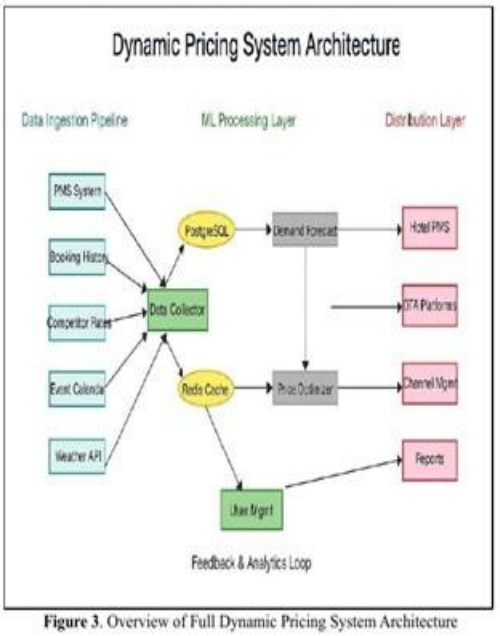

Figure 3. Overview of Full Dynamic Pricing System Architecture

Data Ingestion Pipeline

This pipeline is responsible for collecting, processing, and moving raw data from various sources to a destination[10].

1. PMS System (Property Management System)
   - Provides real-time room inventory, occupancy levels, and booking patterns[6][10].
   - Acts as the hotel's internal data source for operational and pricing decisions[1][7].
2. Booking History
   - Historical records of past bookings including dates, prices, and customer segments[17][18].
   - Used to identify seasonality, trends, and customer behaviour.
3. Competitor Rates
   - Data on competitor hotel prices from OTAs or web scraping tools[14][15].
   - Helps maintain competitive pricing strategies.
4. Event Calendar
   - Local events, holidays, or [4]festivals that influence room demand[17].
   - The system uses this to forecast demand spikes.
5. Weather API
   - Weather conditions affect travel and hotel bookings.

- Integrated through APIs to improve demand prediction accuracy.

6. OTA Data (Online Travel Agency Data)
   - Data from booking platforms like Booking.com or Expedia.
   - Provides real-time booking trends, cancellations, and availability.

ML Processing Layer

This layer processes and analyses the data using machine learning and analytics models[7][14].

1. Data Collection
   - Aggregates and cleans all data from multiple sources.
   - Performs preprocessing like normalization, deduplication, and transformation.
   - Acts as the data pipeline's entry point for all analytics modules[11].
2. PostgreSQL
   - Centralized database for structured data storage.
   - Stores historical data, pricing rules, and model outputs for efficient retrieval[11][7].
3. Redis Cache
   - Used for fast data access and real-time computations.
   - Helps speed up machine learning inferences and API responses.[7][3]
4. Demand Forecast
   - Machine learning model predicts future demand for rooms.
   - Inputs include booking[4] trends, events, competitor data, and weather[1][8].
   - Techniques: ARIMA, Random Forest, LSTM, or regression-based models.
5. Price Optimizer
   - Uses outputs from the Demand Forecast to[19] calculate optimal room prices[7].
   - Considers factors like demand elasticity, inventory, and competitor rates[5][17].
   - Objective: Maximize revenue while maintaining occupancy.
6. Analytics
   - Aggregates outputs into interpretable metrics like ADR (Average Daily Rate),RevPAR (Revenue Per Available Room), and occupancy[8].
   - Generates dashboards and visual insights for hotel managers.

Distribution Layer

This layer is responsible for integrating the system with hotel and market distribution platforms.

1. API Gateway
   - Central hub for communication between the pricing engine and external systems[13].
   - Handles authentication, rate updates, and synchronization across platforms.
2. Hotel PMS
   - Receives updated room prices automatically from the system.
   - Ensures consistent pricing across hotel operations.
3. OTA Platforms (Online Travel Agencies)
   - Platforms like Expedia, MakeMyTrip, or Booking.com.
   - Updated automatically via the API Gateway to reflect optimized prices[7].
4. Channel Management
   - Manages distribution across multiple OTAs and booking channels.
   - Prevents overbooking and maintains rate parity.
5. Reports
   - Generates detailed reports on performance metrics, pricing changes, and trends [8][18].
   - Helps in decision-making and strategy refinement.

IV. ANALYSIS

The analytical backbone consists of forecasting, elasticity modeling, and constrained optimization, each governed by explicit performance criteria and continuous monitoring to ensure resilience under data and concept drift. Forecasting segments series by room type, channel, and lead-time cohorts to reflect heterogeneous booking curves, employing SARIMA/SARIMAX for strong seasonality and boosted trees for nonlinear interactions among event proximity, weather deviations, competitor deltas, and search intensity; probabilistic outputs support risk-aware pricing and are validated through Backtesting and calibration checks in a champion–challenger framework. Elasticity analysis estimates demand sensitivity to price across contexts—weekday vs. weekend, event vs. non-event, direct vs. OTA—and measures cross-price elasticity where competitor moves and OTA ranking impacts alter effective sensitivity; this composite elasticity steers the optimizer away from locally optimal but strategically harmful price points that can reduce visibility and long-run RevPAR [15][16].

The optimization process maximizes expected RevPAR or contribution margin by evaluating candidate prices against demand forecasts and elasticity constraints (e.g., parity, inventory, and distribution costs). Scenario analysis tests robustness to market shocks such as events, competitor actions, and weather changes. Operational analysis defines

system performance targets like real-time responsiveness, reliability, and rollback agility, using observability data to fine-tune caching and safety controls. Ablation studies measure the impact of external data and model variants, while reinforcement learning policies are benchmarked against heuristic optimizers under strict safety conditions. Business outcomes are validated through A/B testing, measuring financial gains (RevPAR, ADR, occupancy, and margin improvements) and responsiveness [18] during high-demand events [20].

V. MAJOR FINDINGS

After studying research papers from previous editions, the researcher recognized the main issues with Dynamic Pricing in Hotel System. Many problems and difficulties have surfaced in the field recently. The following are some of the present problems and difficulties in dynamic pricing[18][13]:

1. Revenue Uplift through Forecast-and-Optimize Loops

The integrated use of forecasting and optimization models consistently demonstrates measurable improvements in Revenue per Available Room (RevPAR) and overall profitability. When forecasting and pricing operate in a closed feedback loop, the system learns from actual booking [12][4]patterns, automatically refining parameters and improving accuracy with every cycle. The magnitude of revenue uplift is primarily driven by four factors—signal diversity, data refresh cadence, integration fidelity, and governance mechanisms. Broader signal coverage (events, weather, competitor data) increases contextual awareness, while high-frequency refreshes ensure responsiveness to market dynamics. Tight integration across PMS and OTA layers reduces latency, and robust governance policies prevent instability or price parity violations across distribution channels [4][9].

2. Steady-State Market Performance

In stable or predictable markets, the system typically achieves mid-single-digit percentage increases in RevPAR, reflecting reduced mispricing, improved elasticity estimation, and optimal balance between rate and occupancy. These steady-state gains arise because the algorithm minimizes revenue leakage from manual pricing errors and better aligns with customer willingness-to-pay. Furthermore, by embedding distribution cost awareness into optimization, the model avoids the pitfall of pursuing Average Daily Rate (ADR) growth in isolation. Instead, it aligns pricing with overall occupancy optimization and contribution margin, ensuring sustainable profitability rather than superficial price hikes [14].

3. Compression Event Advantage

During high-demand or compression events—such as festivals, conferences, or sudden demand surges—the system's early-detection mechanisms and real-time parity-safe escalation enable rapid adjustments before competitors react. This proactive pricing strategy produces double-digit RevPAR gains, demonstrating the value of near-real-time market responsiveness. Probabilistic forecasting models assess demand uncertainty, while pacing guardrails and safety constraints prevent overreactions that could damage brand trust. By ensuring consistent rate distribution across all channels and reducing stale-rate propagation, the system protects the property from lost revenue opportunities during critical high-traffic windows [13].

4. Hybrid Forecasting Superiority

Empirical analysis confirms that hybrid forecasting frameworks—combining time-series models (e.g., ARIMA, SARIMA) with boosted ensemble learners (e.g., XGBoost, LightGBM)—significantly outperform single-model baselines. The hybrid approach benefits from both worlds: seasonal models provide long-term structural reliability, while boosted trees adapt to short-term nonlinear effects from external drivers such as event proximity, weather anomalies, and competitor promotions. This multi-model ensemble architecture enhances robustness, reduces variance, and maintains interpretability. The modular nature of this design allows dynamic weighting of models depending on context, ensuring stable accuracy even under volatile market conditions [2].

5. Model Interpretability and Drift Monitoring

To ensure stakeholder confidence and regulatory compliance, interpretability is central to the system's ML deployment. Feature attribution tools like SHAP values and LIME explain which factors—event proximity, competitor pricing, booking lead time, or weather—contributed to each pricing recommendation. This transparency helps hotel managers trust algorithmic outputs and facilitates auditability. At the same time, model drift detection keeps an eye on how performance changes over time—things like shifts in the market, changing data patterns, or features that just stop mattering. When drift hits a certain point, automated retraining kicks in to keep forecasts sharp and stop the system's accuracy from sliding off track.[4][10]

6. Elasticity Segmentation and Cross-Channel Feedback

Pricing elasticity isn't the same everywhere. It changes depending on where customers book (your own site or an OTA) and when they book (weeks in advance or last minute). By breaking out elasticity models by channel and lead time, you get more accurate price adjustments and can fine-tune strategies for different customer types. Plus, when you watch how rate changes on one channel affect others, you avoid accidentally hurting your demand or visibility elsewhere. And if you factor in OTA ranking algorithms, you're less likely to chase short-term gains that end up hurting your long-term visibility. All together, this turns elasticity from a simple economic idea into a real-world, multi-channel feedback

engine—and that makes your model work a lot better in practice.[5][17].

7. Integration Quality and Operational Reliability

Integration matters just as much as the machine learning model itself. If your pricing engine, PMS, OTAs, and APIs don't talk smoothly to each other, you'll run into problems fast. Solid two-way sync, smart error handling, and reliable rate reconciliation make sure updates flow quickly and accurately. Always-on parity checks catch any mismatches before they spook customers or cause compliance headaches. And if something goes wrong, rollback tools let you jump back to a safe place without missing a beat. These improvements turn theoretical pricing smarts into real revenue gains—so they're absolutely essential for making the project work.[2]

8. Pragmatic Adoption Path

Jumping straight from manual to AI-powered pricing is a big leap. Take it one step at a time. Start by modeling with historical data and manual checks. Then, add outside signals—think competitor rates and local events. Next, layer in elasticity-driven optimization. Only after you've built some trust and seen results should you go for full automation. But always keep people in the loop. Each step gives you proof it's working, builds confidence, and keeps everything transparent before letting the system fly solo. This way, you avoid chaos and actually get your team on board[14].

9. Governance and Evidence-Based Dashboards

Good governance comes down to three things: transparency, traceability, and proof. Interactive dashboards tie together the numbers that matter—ADR, RevPAR, occupancy—right alongside your model's confidence, forecast accuracy, and the logic behind its decisions. You get built-in A/B testing and experiment tracking, so you can see what's working and why. By tying every algorithmic call directly to real financial results [7][12], you build trust with stakeholders and hold everyone accountable. That's how you make sure the AI is actually driving value, not just spinning its wheels.[11].

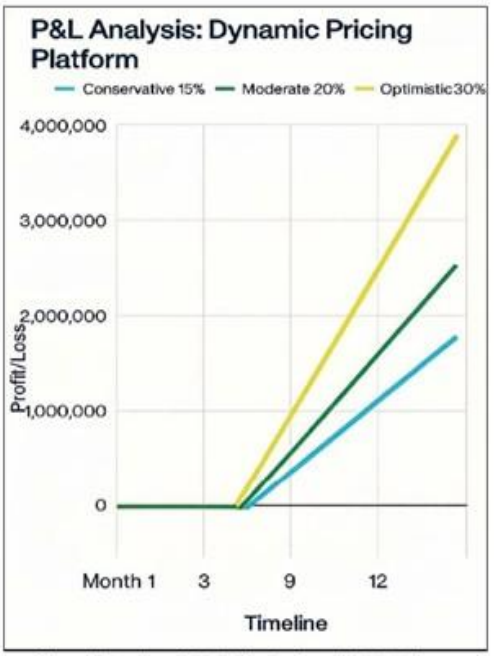

Figure 4: Overview of Profit & Loss in Dynamic Pricing System

VI. CONCLUSION

Picture an AI-driven pricing system that actually makes a difference for hotels. It pulls in real data, predicts demand, understands how different segments react to price changes, and sticks to the rules you set—all while making sure your PMS and OTA connections stay strong and your dashboards stay sharp. The result? Higher ADR, better RevPAR, and stronger occupancy, even when the market's a mess. You don't have to worry about losing parity, stability, or clarity. This isn't just about fancy analytics. It's about making smart tech actually work in the real world. The system uses machine learning you can actually understand, so people trust it. It constantly tests itself, checks for drift, and makes sure compliance and parity rules don't slip. Humans stay in the loop, so there's always accountability. It doesn't just talk a big game, either. It measures how well it predicts and, more importantly, how much real money it makes for you. A/B testing and simulation drills connect the tech with your bottom line. But here's the thing: the real magic depends on your data quality, how often you update, how well you integrate, and how tightly you manage governance. Go slow and steady with changes—roll them out bit by bit. That approach always leads to better results, and your team's much more likely to get on board. Jumping in all at once? That's just asking for trouble.

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APPENDIX -D
POSTER

DYNAMIC PRICING SYSTEM FOR HOTELS USING ML AND DATA ANALYTICS

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(AIML& DS)
SHRI RAMSWAROOP MEMORIAL COLLEGE OF ENGINEERING AND MANAGEMENT, LUCKNOW

PROJECT DONE BY : (GROUP - BTP26/AL/27)
SHREYANSH VERMA(2201221530052)
VIRENDRA VIKRAM SINGH(2101221530060)

UNDER THE GUIDANCE OF :
Er. HIMANSHI SINGH

INTRODUCTION

This project is a real-time dynamic pricing intelligence system built with Python, FastAPI, and Machine Learning. It integrates live occupancy data with predictive models (Random Forest, LSTM) to generate demand-aware room rate recommendations. The system translates historical booking patterns and external demand signals into optimised pricing decisions for hotel revenue managers. It targets key hospitality KPIs including ADR, RevPAR, and Occupancy Rate through an automated, data-driven pricing engine

OBJECTIVES

- Develop a real-time hotel room demand forecasting system using historical booking and occupancy data.
- .Train machine learning models (Random Forest, LSTM) for accurate short and long-range demand prediction.
- Implement a dynamic pricing engine that automatically adjusts room rates based on demand signals.
- Design an interactive dashboard visualising KPIs such as ADR, RevPAR, and Occupancy Rate.

METHODOLOGY

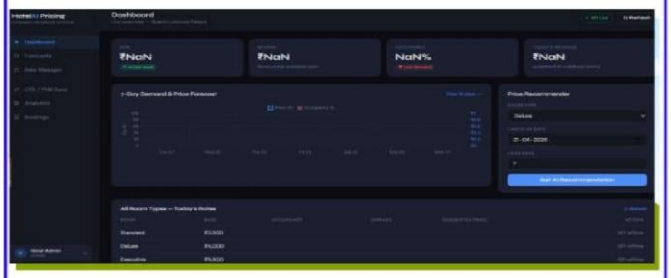
- implemention of 3-tier system utilising a Python/FastAPI backend with SQLite/PostgreSQL database.
- Integrated demand forecasting pipeline with a built-in mock-data fallback for offline operation.
- Engineered a dynamic pricing engine applying demand-based multipliers with surge pricing tiers.
- Built OTA adapter modules for Booking.com, Expedia Rapid, and Mews PMS rate synchronisation.

SYSTEM ARCHITECTURE

MODULE DESCRIPTION

1. Data & Booking Engine: Collects real-time booking data with seamless SQLite/PostgreSQL storage and offline mock-data fallback.
2. Predictive ML Core: Leverages Random Forest and LSTM models for short and long-range occupancy demand forecasting.
3. Dynamic Pricing Engine: Converts occupancy predictions into optimised room rates using demand-tier multipliers and surge pricing logic.
4. OTA / PMS Integration: Pushes recommended rates to Booking.com, Expedia, and Mews PMS simultaneously via API adapters.
5. Revenue Dashboard: Visualises live KPIs (ADR, RevPAR, Occupancy) with 7–30 day forecast charts and price history logs

VISUALIZATION



ADVANTAGES

- Translates raw occupancy forecasts into actionable, revenue-maximising room pricing decisions in real time.
- Guarantees 100% system functionality without live OTA credentials via intelligent simulation mode fallback.
- Highly modular architecture that runs entirely on standard hardware with zero licensing costs.

CONCLUSION

- Implemented a 3-tier system utilising a Python/FastAPI backend and SQLite relational database.
- Integrated Random Forest and LSTM models delivering accurate short and long-range demand forecasts.
- Engineered a surge-aware dynamic pricing engine respecting per-room minimum and maximum price bounds.

FUTURE WORK

- Advanced AI: Integrate Transformer-based models for continuous online learning and probabilistic demand forecasting.
- Live OTA Data: Connect real-time competitor rate scraping to incorporate market-aware pricing signals.
- Mobile App: Launch native iOS/Android versions with push notifications for demand surge alerts and pricing approvals.

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